Computer Science - Class 9 (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

12 chapters — 345 exercises

Chapter 1: Introduction to Systems

Multiple Choice Questions (MCQs)

1. What is the primary function of a system?

(a) To work independently

(b) To achieve a common goal ✓

(c) To create new systems

(d) To provide entertainment

Correct answer: To achieve a common goal

Multiple Choice Questions (MCQs)

2. What is one of the fundamental concepts of any system?

(a) Its size

(b) Its objective ✓

(c) Its age

(d) Its price

Correct answer: Its objective

Multiple Choice Questions (MCQs)

3. What is an example of a simple system?

(a) A human body

(b) A computer network

(c) A thermostat regulating temperature ✓

(d) The Internet

Correct answer: A thermostat regulating temperature

Multiple Choice Questions (MCQs)

4. What type of environment remains unchanged unless the system provides an output?

(a) Dynamic

(b) Static ✓

(c) Deterministic

(d) Non-deterministic

Correct answer: Static

Multiple Choice Questions (MCQs)

5. What are the basic components of a system?

(a) Users, hardware, software

(b) Objectives, components, environment, communication ✓

(c) Inputs, outputs, processes

(d) Sensors, actuators, controllers

Correct answer: Objectives, components, environment, communication

Multiple Choice Questions (MCQs)

6. What concept does the theory of systems aim to understand?

(a) Hardware design

(b) System interactions and development over time ✓

(c) Software applications

(d) Network security

Correct answer: System interactions and development over time

Multiple Choice Questions (MCQs)

7. What role does the Operating System (OS) play in a computer?

(a) It performs calculations and executes instructions

(b) It temporarily stores data and instructions for the CPU

(c) It receives input from interface components and decides what to do with it ✓

(d) It provides long-term storage of data and software

Correct answer: It receives input from interface components and decides what to do with it

Multiple Choice Questions (MCQs)

8. Which of the following describes the Von Neumann architecture's main characteristic?

(a) Separate memory for data and instructions

(b) Parallel execution of instructions

(c) Single memory store for both program instructions and data ✓

(d) Multiple CPUs for different tasks

Correct answer: Single memory store for both program instructions and data

Multiple Choice Questions (MCQs)

9. What is a disadvantage of the Von Neumann architecture?

(a) Complex design due to separate memory spaces

(b) Difficult to modify programs stored in memory

(c) Bottleneck due to single memory space for instructions and data ✓

(d) Lack of flexibility in executing instructions

Correct answer: Bottleneck due to single memory space for instructions and data

Multiple Choice Questions (MCQs)

10. Which of the following transports data inside a computer among different components?

(a) Control Unit

(b) System Bus ✓

(c) Memory

(d) Processor

Correct answer: System Bus

Short Questions

1. Define a system. What are its basic components?

A system is an organized set of components that are coordinated to perform a designated function: every part is related to the others, and the parts work together to achieve an outcome. A system is described by four basic components: its objective (the purpose or goal it must fulfil), its components (the parts that make it up), its environment (everything outside the system that interacts with it), and communication (how the components interact with each other). A system can be simple, like a thermostat, or complex, like the human body or a computer network.

Short Questions

2. Differentiate between natural and artificial systems.

Natural systems exist in nature and operate independently of human involvement, governed by natural laws; examples are physical systems (planets, stars, galaxies), chemical systems (water formed when hydrogen bonds with oxygen), biological systems (living organisms) and psychological systems (thoughts and emotions). Artificial systems are created and developed by people to fulfil certain functions or solve problems, for example knowledge systems, engineering systems and social systems; they range from something as small as a wheel to something as large as the United Nations.

Short Questions

3. Describe the main components of a computer system.

A computer system has three groups of components. Interface components are the fundamental parts for user interaction: input devices such as the keyboard and mouse, and output devices such as monitors and printers. Processing components are the CPU, which performs computations and executes

commands; RAM, the transient storage holding data and instructions for the CPU; storage (hard drive or SSD), the permanent storage for data and software; the operating system, which receives information from the interface components and decides what action to take; and application software, executed by the operating system to perform specific tasks. Communication components are the physical elements linking the parts: the motherboard, the main circuit board, and the system bus (data bus, address bus and control bus).

Short Questions

4. List and describe the types of computing systems.

The book lists four types of computing systems. (1) Computers: a complex system designed to process data and perform tasks according to a set of instructions. (2) Software systems: a structured set of hardware and software components designed for data processing, including system software (operating systems and utilities) and application software (word processors, web browsers, games). (3) Computer networks: systems that connect multiple computers and devices so they can share resources and exchange information efficiently. (4) The Internet: a vast, complex system connecting many networks worldwide, whose primary objective is communication and data exchange between computers and users globally.

Short Questions

5. What are the main components of the Von Neumann architecture?

The Von Neumann architecture, named after John von Neumann (1940s), has four primary components. Memory holds both the input data and the instructions (program) needed for CPU processing; for example, a program is loaded into RAM for faster execution than from the hard disk. The Central Processing Unit (CPU) performs arithmetic and executes commands from memory; it has two

parts, the Arithmetic Logic Unit (ALU) and the Control Unit (CU). Input devices (keyboard, mouse, microphone) let users enter data and instructions. Output devices (monitor, printer) present the results of processing.

Short Questions

6. What is the Von Neumann computer architecture? List its key components.

The Von Neumann architecture is a computer design model in which the hardware has four primary components: memory, the Central Processing Unit (CPU), input mechanisms and output mechanisms, all linked by a system bus. Its key characteristics are a single memory store (program instructions and data share the same memory space), sequential execution (instructions are processed one after another in the order written), and the stored-program concept (programs are stored in memory and can be changed by the computer, so updating software replaces old instructions with new ones in memory). The CPU itself contains the Control Unit (CU), which governs the CPU by instructing the ALU and memory, and the Arithmetic Logic Unit (ALU), which performs mathematical and logical operations.

Short Questions

7. What are the four main steps in the Von Neumann architecture's instruction cycle?

The four steps are fetching, decoding, executing and storing. (1) Fetching: the CPU retrieves an instruction from memory; the Program Counter (PC) holds the address of the next instruction, and the instruction at that address is placed in the Instruction Register (IR). (2) Decoding: the Control Unit (CU) decodes the instruction, reading its operation code (opcode) to determine the required action, procedures and data. (3) Execution: the CPU carries out the instruction; computations are done by the Arithmetic Logic Unit (ALU), while data transfers

between locations are managed by the CU. (4) Storing: the outcome is returned to memory or sent to an output device such as a display.

Short Questions

8. What is the Von Neumann bottleneck?

The Von Neumann bottleneck occurs when a single memory area limits the CPU's ability to retrieve instructions and data quickly. Because both program instructions and data share one memory space and one pathway (the system bus), the CPU must fetch them through the same channel, and this shared path becomes a choke point that slows the CPU down. It is listed in the book as a disadvantage of the architecture, alongside the security risk that one program can alter another program's instructions because data and instructions sit in the same area.

Short Questions

9. What is a key advantage of the Von Neumann architecture?

Its key advantages are simplified design and flexibility. Combining instructions and data into a single memory area simplifies the architecture, since one memory and one bus serve both purposes instead of two separate systems. It is also flexible: programs can easily be changed simply by changing the contents of memory, so new software replaces old instructions without redesigning the hardware. These two advantages explain why the model became the central design for most computers despite the bottleneck limitation.

Short Questions

10. What are the three main requirements for a computing system to function?

The book names three basic requisites for running a computing system. Hardware: the tangible components, including the CPU, RAM, storage devices

and input/output devices. Software: the collection of instructions that dictate what the hardware must do, in two categories — system software (operating systems and utilities that manage the computer's resources, such as Windows, macOS and Linux) and application software (programs built for specific user functions, such as word processors, web browsers and games). Electricity: the power source that enables the hardware to function; without electricity the hardware components cannot function and the computing system will not operate.

Long Questions

1. Define and describe the concept of a system. Explain the fundamental components, objectives, environment, and methods of communication within a system.

A system is an organized set of components coordinated to perform a designated function, so that all parts are related to each other and their combined functioning enhances the operation of the whole; a car is the book's example, where engine, wheels and brakes each do a unique task but together make the car move. Its fundamental components are its objective, its components, its environment and communication. Every system has a purpose or goal: common objectives are information processing (collecting, storing, processing and distributing information, as a computer system or the human brain does), supporting other systems (providing a platform or infrastructure, as a cell phone does for applications, or the Sun does by providing energy for life), and achieving specific goals (completing a task, as a thermostat maintaining a set temperature or a car engine converting fuel into mechanical energy efficiently). Components are the building blocks, each with a specific role contributing to overall functionality. The environment is everything external to the system that interacts with it and affects its operation through inputs and

outputs; its properties include static versus dynamic (a static environment stays unchanged unless the system produces an output, while a dynamic environment can change independently of the system) and deterministic versus non-deterministic (in a deterministic system the impact of an output is fully known and certain, while in a non-deterministic system it involves uncertainty, randomness or probability). Communication among components is what lets them work together in an organized, smooth way to achieve the objective — for example, in a computing system the CPU communicates with memory to fetch and store data, and in a biological system the brain sends signals to muscles to initiate movement.

Long Questions

2. Differentiate between natural and artificial systems. Discuss their characteristics, functions, and purposes with relevant examples.

Natural systems exist in nature and operate independently of human involvement, governed by natural laws and processes, and range from tiny objects like atoms and cells to huge ones like forests, oceans and the cosmos. They divide into physical systems (composed of physical components governed by physics, from sub-atomic particles and atoms to planets, stars, galaxies and the cosmos; hydrogen gas forms when an electron, proton and neutron combine under physical laws), chemical systems (substances and their interactions, transformations and reactions governed by chemistry; water forms when hydrogen atoms bond with oxygen atoms), biological systems (living organisms and their interactions, governed by growth, reproduction and metabolism; molecules organize into cells, tissues, organs and organisms), and psychological systems (the mind and behaviour — thoughts, emotions and mental processes governed by psychology — emerging from the brain's physical and chemical processes). Artificial systems are created and developed by people to fulfil

certain functions or solve problems, deliberately designed to perform a task, improve process efficiency and provide solutions across sectors, and they reinforce productivity, solve complex problems and improve well-being. They divide into knowledge systems (developed to capture, process, store, retrieve and manage information for decision-making, learning and problem-solving — mathematics, logic, databases such as MySQL and MongoDB, and information management systems), engineering systems (complex frameworks or devices applying engineering concepts — civil systems such as houses, roads and bridges like Attock Bridge; mechanical systems such as a robotic arm on an assembly line; chemical systems such as a water treatment plant; electrical systems such as home automation controlled by a smartphone app; and software engineering systems such as an online library tool), and social systems (structured frameworks for social interaction, governance and communal endeavour, aiming to maintain order, provide services and facilitate connections — academic institutions such as schools and universities including Punjab University, governments, and organizations such as Apple or the Edhi Foundation).

Long Questions

3. Examine the relationship between systems and different branches of science, including natural science, design science, and computer science. How do these branches utilize system theory to understand and improve their respective fields? Provide specific examples to support your analysis.

Systems theory is the branch of science that deals with complicated structures in living organisms and in human society, giving a way to interpret the world through different perspectives: how systems and sub-systems operate, how they are integrated, how they grow and how they change with time. Science divides into natural science and design science, which both study systems but approach

them differently. Natural science is descriptive: scientists study existing natural systems to understand their workings, following the empirical cycle of observation, question, hypothesis, experiment, analysis and conclusion — for example, studying a forest ecosystem to understand how species interact. Design science is prescriptive: researchers design and create artifacts (tools, systems, methods) to achieve specific goals, following the regulative cycle of problem investigation/identification, solution design, solution implementation and solution evaluation — for example, developing new software to manage forest data and improve conservation. Computer science uses both methods: its natural-science side finds the basic rules controlling how computer systems work, for instance analyzing existing algorithms such as QuickSort and MergeSort to understand their efficiency and limitations with different data; its design-science side creates and improves computer tools and systems, for instance developing a new programming language that makes it easier to write secure programs, or a more efficient database system handling larger data faster with fewer errors. So system theory lets natural science describe systems as they are, design science prescribe systems as they should be, and computer science do both.

Long Questions

4. Explore the different types of computing systems such as computers, software systems, computer networks, and the internet.

A computing system is a structured set of hardware and software components specifically designed for data processing and various operations, from simple tools like calculators to complex networks of linked computers, whose basic task is to execute programs and manage data for problem solving, process control and communication. (1) Computers: a complex system designed to process data and perform tasks according to instructions. Its main objective is to perform

computations, process data and execute tasks efficiently — for example, a personal computer running word processors, web browsers and games. Its components are interface components (keyboard, mouse, monitors, printers), processing components (CPU, RAM, storage, operating system, application software) and communication components (motherboard and system bus). (2) Software systems: the structured combination of system software and application software. System software is the basic software that helps a computer run and manage its hardware and software resources (operating systems and utilities); application software is designed to help users perform specific tasks (word processors, browsers, games). (3) Computer networks: systems connecting multiple computers and devices to exchange resources and information efficiently. Their objectives are resource sharing (files, printers, internet access within an office), communication (efficient exchange between devices and users) and data management (easy management and collaboration). Their components are networking hardware (routers transmitting data packets between networks, switches connecting devices, network cables as the physical medium) and network software (protocols such as TCP/IP, and network operating systems such as Windows Server). Types are Local Area Networks (LANs, one area such as a building or school) and Wide Area Networks (WANs, larger regions such as cities, nations and continents — the Internet itself). (4) The Internet: a vast, complex system connecting multiple networks worldwide — private, public, academic, business and government — whose primary objective is communication and data exchange between computers and users globally. Its core protocols are TCP/IP (governing data transmission), UDP (faster but less reliable), FTP (transferring files between computers) and POP (retrieving emails from a server). When a user requests a web page, several Internet components work together to display it on screen, and the Internet operates in a diverse, dynamic environment of homes, offices, data centers and

mobile networks that influences its design, security and performance.

Long Questions

5. Describe the main characteristics of a computer as a system, including its objectives, components, and interactions among these components.

A computer is a complex system designed to process data and perform tasks according to a set of instructions. Its main objective is to perform computations, process data and execute different tasks efficiently, for example running word processors, web browsers and games through various computational processes. Its components fall into three groups: interface components (input devices such as keyboard and mouse for user interaction, and output devices such as monitors and printers that present results), processing components (the CPU for computations and commands; RAM as transient storage for data and instructions for the CPU; storage such as a hard drive or SSD as permanent storage; the operating system, which receives information from interface components and determines the appropriate action; and application software executed by the operating system for specific tasks), and communication components (the motherboard as the primary circuit board interconnecting all components, and the system bus — data bus, address bus and control bus — carrying data, addresses and control signals concurrently). The components interact seamlessly: when you open a file, for example by double-clicking 'report.docx' on the desktop, the mouse or keyboard sends a signal to the operating system (through the USB connection), the operating system locates the file in storage, loads it into RAM, the CPU processes the open request, and the result is displayed on the monitor. The computer also interacts with its environment: external devices such as the power supply (electrical power to work), the network (connection to other systems and the Internet) and peripherals (printers, scanners, external discs) — for example, a user typing on the keyboard

is processed and displayed as text on screen, the computer sends and receives data over the Internet to browse or download, and it relies on a stable power supply to function correctly.

Long Questions

6. Explain the Von Neumann architecture of a computer. Include a discussion on the main components, their functions, and the step-by-step process of how the architecture operates.

The Von Neumann architecture, the John von Neumann model developed in the 1940s, is a computer paradigm in which the hardware has four primary components: memory, the Central Processing Unit (CPU), input mechanisms and output mechanisms. Memory contains both input data and the instructions (program) required for CPU processing — when a program starts it is loaded into RAM for faster execution than from the hard disk. The CPU performs arithmetic (addition, subtraction) and executes commands from memory; it has two parts, the Arithmetic Logic Unit (ALU), which performs mathematical and logical operations, and the Control Unit (CU), which governs the CPU by instructing the ALU and memory so tasks execute properly and on time (in the book's calculator example, the ALU handles the numbers for 2 + 2 while the CU supervises the whole procedure). Input devices (keyboard, mouse, microphone) enter data and instructions, and output devices (monitor, printer) present the outcomes. A system bus links them: the data bus transports data, the address bus maintains destination information, and the control bus transports control signals. Operation follows four steps. Fetching: the CPU retrieves an instruction from memory, using the Program Counter (PC) address, and places it in the Instruction Register (IR). Decoding: the CU decodes the instruction's operation code (opcode) to determine the action, procedures and data. Execution: the CPU carries out the instruction, with the ALU doing computations and the CU managing data

transfers. Storing: the outcome is returned to memory or sent to an output device such as a display. Its characteristics are a single memory store (code and data, e.g. a game's code and its scores and player positions, share the same RAM), sequential execution (instructions run one after another in written order), and the stored-program concept (programs in memory can be changed by the computer, so a software update replaces old instructions with new ones). Its advantages are simplified design and flexibility; its disadvantages are the Von Neumann bottleneck (one memory area limiting how fast the CPU retrieves instructions and data) and security risks (one program altering another's instructions in shared memory).

Long Questions

7. Provide a detailed explanation of how a computer interacts with its environment. Include examples of user input, network communication, and power supply.

A computer system constantly interacts with its environment through inputs and outputs, and its environment includes any external devices that interact with it: the power supply, the network and peripherals. For user input: a user types on the keyboard, and the computer processes the input to display text on the screen — the keyboard sends a signal (for example through a USB connection) to the operating system, which interprets the keystrokes, and the application shows the characters on the monitor. The book's weather-monitoring example shows the same pattern in another system: sensors receive data from the environment and the system provides current status and forecasts to users; and in a biological system, animals interact with plants and other animals, forming a food chain. For network communication: the computer sends and receives data over the Internet to browse websites or download files — the network connects the computer to other systems and the Internet, and when a user requests a web

page, several Internet components work together to display its contents on the user's screen. For power supply: the computer relies on a stable power supply to function correctly, since electricity is the power source enabling the hardware to function; without electricity the hardware cannot function and the computing system will not operate. Peripherals extend the same interaction: printers, scanners and external discs expand what the computer can do with its environment, receiving its outputs or feeding it new inputs.

Long Questions

8. Describe the process of retrieving and displaying a file using a computer, based on the interactions among different components. Provide a step-by-step explanation of how input is processed, data is transferred, and results are displayed on the screen.

The book illustrates component interaction with opening a file, and the same steps describe retrieving and displaying any file. Step 1 — User action or input: you double-click a file icon using the mouse, or press a key combination to open a file, for example double-clicking a document named 'report.docx' on the desktop. Step 2 — Input device: the mouse or keyboard sends a signal indicating that you want to open the file, for example the mouse sends sensory input to the computer's operating system through the USB connection. Step 3 — Operating system and processing: the operating system receives the information from the interface component and determines the appropriate action; it locates the file in permanent storage (hard drive or SSD) and loads it into RAM, the transient storage that holds data and instructions for the CPU. Step 4 — Data transfer: the system bus moves the data between components — the data bus transports the file's data, the address bus carries the destination information, and the control bus carries the control signals from the CPU — with the motherboard interconnecting all components. Step 5 — Processing: the CPU executes the

instructions for opening the file, with the Control Unit supervising the procedure and the Arithmetic Logic Unit handling any computations. Step 6 — Display: the outcome is sent to an output device, so the file's contents appear on the monitor for the user to read and work with. If the file is later saved, the updated data travels back along the bus to storage for future processing.

Chapter 2: Number Systems

Multiple Choice Questions (MCQs)

1. What does ASCII stand for?

(a) American Standard Code for Information Interchange ✓

(b) Advanced Standard Code for Information Interchange

(c) American Standard Communication for Information Interchange

(d) Advanced Standard Communication for Information Interchange

Correct answer: American Standard Code for Information Interchange

Multiple Choice Questions (MCQs)

2. Which of the following numbers is a valid binary number?

(a) 1101102

(b) 11011 ✓

(c) 110.11

(d) 1101A

Correct answer: 11011

Multiple Choice Questions (MCQs)

3. How many bits are used in the standard ASCII encoding?

(a) 7 bits ✓

(b) 8 bits

(c) 16 bits

(d) 32 bits

Correct answer: 7 bits

Multiple Choice Questions (MCQs)

4. Which of the following is a key advantage of Unicode over ASCII?

(a) It uses fewer bits per character

(b) It can represent characters from many different languages ✓

(c) It is backward compatible with binary

(d) It is specific to the English language

Correct answer: It can represent characters from many different languages

Multiple Choice Questions (MCQs)

5. How many bytes are used to store a typical integer?

(a) 1 byte

(b) 2 bytes

(c) 4 bytes ✓

(d) 8 bytes

Correct answer: 4 bytes

Multiple Choice Questions (MCQs)

6. What is the primary difference between signed and unsigned integers?

(a) Unsigned integers cannot be negative ✓

(b) Signed integers have a larger range

(c) Unsigned integers are stored in floating-point format

(d) Signed integers are only used for positive numbers

Correct answer: Unsigned integers cannot be negative

Multiple Choice Questions (MCQs)

7. In the single precision, how many bits are used for the exponent?

(a) 23 bits

(b) 8 bits ✓

(c) 11 bits

(d) 52 bits

Correct answer: 8 bits

Multiple Choice Questions (MCQs)

8. What is the approximate range of values for single-precision floating-point numbers?

(a) 1.4 x 10^-45 to 3.4 x 10^38

(b) 1.4 x 10^-38 to 3.4 x 10^45

(c) 4.9 x 10^-324 to 1.8 x 10^308

(d) 4.9 x 10^-308 to 1.8 x 10^324

Correct answer: 1.4 x 10^-45 to 3.4 x 10^38

Multiple Choice Questions (MCQs)

9. What are the tiny dots that make up an image called?

(a) Pixels

(b) Bits

(c) Bytes

(d) Nodes

Correct answer: Pixels

Multiple Choice Questions (MCQs)

10. In an RGB color model, what does RGB stand for?

(a) Red, Green, Blue

(b) Red, Gray, Black

(c) Right, Green, Blue

(d) Red, Green, Brown

Correct answer: Red, Green, Blue

Short Questions

1. What is the primary purpose of the ASCII encoding scheme?

ASCII gives every English letter, digit, punctuation mark and control character a

unique number from 0 to 127, so that text can be stored and exchanged as numbers. Because all computers agree on the same table, a file written on one machine reads correctly on another. It is the basic scheme on which later, wider schemes were built.

Short Questions

2. Explain the difference between ASCII and Unicode.

ASCII is a 7-bit scheme with only 128 characters, covering English letters, digits and common symbols, and it cannot represent other languages. Unicode is a universal standard that assigns a unique code point to characters from all the world's writing systems, including Urdu, Arabic, Chinese and symbols. Unicode can be stored in several encodings, such as UTF-8, UTF-16 and UTF-32, while ASCII's characters form the first 128 code points of Unicode.

Short Questions

3. How does Unicode handle characters from different languages?

Unicode gives each character, in every script, one unique code point, regardless of language, so the same system covers English, Urdu, Arabic, Greek, Chinese and emoji side by side. A character is stored by its code point using an encoding like UTF-8, which uses one to four bytes as needed, so no language needs a separate, conflicting table. Software that supports Unicode can therefore display and exchange mixed-language text correctly.

Short Questions

4. What is the range of values for an unsigned 2-byte integer?

A 2-byte (16-bit) unsigned integer has 2^16 = 65,536 possible values. Since unsigned integers cannot be negative, the range is 0 to 65,535. The smallest value uses all zero bits (0000000000000000) and the largest uses all one bits

(1111111111111111).

Short Questions

5. Explain how a negative integer is represented in binary.

Negative integers are normally stored in two's complement form. To represent a negative number, write its positive value in binary, invert every bit (one's complement), and add 1. For example, in 4 bits, +5 is 0101, inverting gives 1010, and adding 1 gives 1011, which is -5. The leftmost bit then acts as the sign bit: 1 means negative, 0 means positive.

Short Questions

6. What is the benefit of using unsigned integers?

When a quantity can never be negative, such as a count, an age or a memory size, an unsigned integer uses the sign bit for value instead of wasting it on a sign. This doubles the largest positive value that fits in the same number of bits: an 8-bit unsigned integer reaches 255 instead of 127. It also makes the meaning clear to the reader: negative values are impossible by design.

Short Questions

7. How does the number of bits affect the range of integer values?

With n bits, 2^n different values can be represented, so every extra bit doubles the range. An 8-bit unsigned integer covers 0 to 255, a 16-bit one covers 0 to 65,535, and a 32-bit one covers 0 to 4,294,967,295. For signed integers the range is split between negative and positive, for example -128 to +127 in 8 bits, but the doubling rule is the same.

Short Questions

8. Why are whole numbers commonly used in computing for quantities that

cannot be negative?

Counts, sizes and positions are naturally whole and never negative, so an unsigned whole number matches the quantity exactly and uses the full bit range for positive values. Whole numbers are also exact in binary, unlike fractions, which may only be approximated, so counting and indexing stay reliable. Using a signed type for such a quantity would waste half the range on impossible negative values.

Short Questions

9. How is the range of floating-point numbers calculated for single precision?

A single-precision float uses 32 bits: 1 sign bit, 8 exponent bits and 23 mantissa (fraction) bits. The exponent, with a bias of 127, can represent powers from about 10^-45 (smallest denormal) up to about 10^38 (largest normal), and the 23-bit mantissa gives roughly 7 decimal digits of precision. The range is therefore approximately 1.4 x 10^-45 to 3.4 x 10^38, positive and negative, as in MCQ 8.

Short Questions

10. Why is it important to understand the limitations of floating-point representation in scientific computing?

Floating-point numbers have limited precision, so many decimal fractions cannot be stored exactly and small rounding errors appear in every operation. In long scientific calculations these errors accumulate and can change results, make two mathematically equal values compare unequal, or overflow to infinity. A programmer who knows the limits chooses suitable precision, avoids careless equality tests on floats, and checks results instead of trusting them blindly.

Long Questions

1. Explain how characters are encoded using Unicode. Provide examples of characters from different languages and their corresponding Unicode code points.

Unicode assigns every character a unique number called a code point, written as U+ followed by hexadecimal digits. Examples: the English letter 'A' is U+0041, the Arabic letter 'Alef' is U+0627, the Euro sign is U+20AC, and the Chinese character for 'middle' is U+4E2D. The code point is then stored with an encoding: UTF-8 uses one byte for ASCII characters and up to four bytes for others, UTF-16 uses two or four bytes, and UTF-32 always uses four bytes. Because the code point is language-independent, the same document can mix English, Urdu and Chinese text without conflict.

Long Questions

2. Describe in detail how integers are stored in computer memory.

Integers are stored as fixed-width binary numbers, typically 1, 2, 4 or 8 bytes, with each bit holding a 0 or 1 and place values doubling from right to left. Unsigned integers use all bits for value, giving 0 to 2^n - 1 for n bits. Signed integers use two's complement: the leftmost bit is the sign (0 positive, 1 negative), positive values are stored directly, and negative values are stored as the two's complement of their magnitude, giving -2^(n-1) to 2^(n-1) - 1. For example, in 8 bits, 25 is 00011001 and -25 is 11100111. Fixed width means a value that exceeds the range overflows and wraps around, which is why choosing the right size matters.

Long Questions

3. Explain the process of converting a decimal integer to its binary representation and vice versa. Include examples of both positive and negative integers.

Decimal to binary: repeatedly divide by 2 and record the remainders, then read them from bottom to top. For 13: 13/2 = 6 remainder 1, 6/2 = 3 remainder 0, 3/2 = 1 remainder 1, 1/2 = 0 remainder 1, giving 1101. Binary to decimal: multiply each bit by its place value (1, 2, 4, 8...) and add: 1101 = 8 + 4 + 0 + 1 = 13. For a negative integer, use two's complement in a fixed width: -5 in 4 bits is the complement of +5 (0101): invert to 1010, add 1 to get 1011. To read 1011 back, note the sign bit 1 (negative), invert to 0100, add 1 to get 0101 = 5, so the value is -5.

Long Questions

4. Perform the following binary arithmetic operations: a. Multiplication of 101 by 11. b. Division of 1100 by 10.

a. Multiplication: 101 (5) x 11 (3). Multiply 101 by each bit of 11: 101 x 1 = 101, and 101 x 1 shifted left = 1010. Adding: 101 + 1010 = 1111. Check in decimal: 5 x 3 = 15, and 1111 = 8 + 4 + 2 + 1 = 15. b. Division: 1100 (12) / 10 (2). 10 goes into 11 once (remainder 1), bring down 0 to make 10, which 10 divides once, and bring down 0 to make 0, giving quotient 110 with remainder 0. Check: 12 / 2 = 6, and 110 = 4 + 2 = 6.

Long Questions

Note: Long Question 5 (numbering)

There is no Long Question 5 printed in the book. On book page 48 (PDF page 51, verified against a render) the Long Questions are numbered 1, 2, 3, 4, 6, 7, 8: the numbering jumps directly from Q4 to Q6. No question has been invented to fill the gap; all printed Long Questions (1-4 and 6-8) are solved above and below.

Long Questions

6. Add the following binary numbers: a) 101 + 110, b) 1100 + 1011.

a. 101 + 110: add column by column from the right: 1 + 0 = 1; 0 + 1 = 1; 1 + 1 = 10, write 0 and carry 1, giving 1011. Check: 5 + 6 = 11, and 1011 = 8 + 2 + 1 = 11. b. 1100 + 1011: 0 + 1 = 1; 0 + 1 = 1; 1 + 0 = 1; 1 + 1 = 10, write 0 and carry 1, giving 10111. Check: 12 + 11 = 23, and 10111 = 16 + 4 + 2 + 1 = 23.

Long Questions

7. Convert the following numbers to 4-bit binary and add them: (a) 7 + (-4), (b) -5 + 3.

(a) 7 in 4 bits is 0111. -4 is the two's complement of 0100: invert to 1011, add 1 to get 1100. Adding: 0111 + 1100 = 10011. The result is 5 bits, so the extra leftmost carry is dropped, leaving 0011 = 3. Check: 7 + (-4) = 3. (b) -5 in 4 bits is 1011 (from Q3) and +3 is 0011. Adding: 1011 + 0011 = 1110. The sign bit is 1, so the result is negative: invert 1110 to 0001, add 1 to get 0010 = 2, so the value is -2. Check: -5 + 3 = -2.

Long Questions

8. Solve the following: (a) 1101 - 0100, (b) 1010 - 0011, (c) 1000 - 0110, (d) 1110 - 100.

(a) 1101 - 0100: borrow where needed: 1 - 0 = 1; 0 - 0 = 0; 1 - 1 = 0; 1 - 0 = 1, giving 1001. Check: 13 - 4 = 9, and 1001 = 8 + 1 = 9. (b) 1010 - 0011: 0 - 1 needs a borrow, giving 1 with borrow; continuing column by column gives 0111. Check: 10 - 3 = 7. (c) 1000 - 0110 = 0010. Check: 8 - 6 = 2. (d) 1110 - 0100 (100 padded to 4 bits) = 1010. Check: 14 - 4 = 10.

Chapter 3: Digital Systems and Logic Design

Multiple-Choice Questions (MCQs)

1. Which of the following Boolean expressions represents the OR operation?

(a) A . B

(b) A + B ✓

(c) A (bar)

(d) A XOR B

Correct answer: A + B

Multiple-Choice Questions (MCQs)

2. What is the dual of the Boolean expression A . 0 = 0?

(a) A + 1 = 1 ✓

(b) A + 0 = A

(c) A . 1 = A

(d) A . 0 = 0

Correct answer: A + 1 = 1

Multiple-Choice Questions (MCQs)

3. Which logic gate outputs true only if both inputs are true?

(a) OR gate

(b) AND gate ✓

(c) XOR gate

(d) NOT gate

Correct answer: AND gate

Multiple-Choice Questions (MCQs)

4. In a half-adder circuit, the carry is generated by which operation?

(a) XOR operation

(b) AND operation

(c) OR operation

(d) NOT operation

Correct answer: AND operation

Multiple-Choice Questions (MCQs)

5. What is the decimal equivalent of the binary number 1101?

(a) 11

(b) 12

(c) 13

(d) 14

Correct answer: 13

Short Questions

1. Define a Boolean function and give an example.

A Boolean function is an expression built from binary variables (0 and 1) using AND, OR and NOT, which returns a single binary value, 0 or 1, for every combination of inputs. An example is F(A, B) = A . B + C, which is 1 when both A and B are 1, or when C is 1, and 0 otherwise. Boolean functions describe exactly what a logic circuit does, so they are the link between a required behaviour and the gates that implement it.

Short Questions

2. What is the significance of the truth table in digital logic?

A truth table lists every possible combination of inputs alongside the output, so it defines a Boolean function completely and without ambiguity. It is used to design circuits (read the output column to find the function), to verify that a circuit works (compare its outputs with the table), and to simplify functions

(group the rows that give 1). For n inputs the table has 2^n rows, which is why it is practical for small functions and the basis for larger design methods like K-maps.

Short Questions

3. Explain the difference between analog and digital signals.

An analog signal varies continuously over time and can take any value in a range, like the smooth wave of a voice in a microphone. A digital signal takes only discrete values, in computers just two levels, 0 and 1, switching sharply between them. Digital signals are easier to store, copy and process without loss, and noise affects them less, because a slightly disturbed 0 or 1 is still read correctly, while the same noise changes an analog value.

Short Questions

4. Describe the function of a NOT gate with its truth table.

A NOT gate, or inverter, has one input and outputs its opposite: it turns 0 into 1 and 1 into 0. Its truth table is: Input A = 0 gives Output = 1; Input A = 1 gives Output = 0. It is written as F = NOT A, or A with a bar over it, and it is used wherever a signal must be inverted, for example to form the complement of a variable in a Boolean expression.

Short Questions

5. What is the purpose of a Karnaugh map in simplifying Boolean expressions?

A Karnaugh map (K-map) is a grid in which each cell stands for one combination of inputs, arranged so that neighbouring cells differ in only one variable. By circling groups of adjacent 1s in powers of two (1, 2, 4, 8...), the variables that change within a group drop out, leaving a shorter product term. This gives a simplified expression with fewer gates than the original, without the trial and

error of algebraic simplification.

Long Questions

1. Explain the usage of Boolean functions in computers.

Boolean functions are the mathematics behind computer hardware: every decision, calculation and control action in a computer is built from AND, OR and NOT operations on binary values. Arithmetic circuits use them to add and compare numbers, memory circuits use them to select and store bits, and control units use them to decide which operation runs next. Software conditions, such as 'if A and B then do X', are the same Boolean logic at program level. In short, Boolean functions turn a required behaviour into a precise rule that gates, and then programs, can follow.

Long Questions

2. Describe how to construct a truth table for a Boolean expression with an example.

Take F(A, B) = A + B (OR). Step 1: list all input combinations: with 2 inputs there are 2^2 = 4 rows: 00, 01, 10, 11. Step 2: add a column for each intermediate result if needed, here none. Step 3: evaluate the expression for each row: 0 + 0 = 0, 0 + 1 = 1, 1 + 0 = 1, 1 + 1 = 1. The finished table is: A B | F; 0 0 | 0; 0 1 | 1; 1 0 | 1; 1 1 | 1. The table proves that F is 1 whenever at least one input is 1, which is the definition of OR.

Long Questions

3. Describe the concept of duality in Boolean algebra and provide an example to illustrate it.

Duality says that every Boolean identity has a dual obtained by swapping AND with OR, and 0 with 1, throughout. If an identity is true, its dual is also true.

Example: the identity A + 0 = A (OR with 0 leaves A unchanged) has the dual A . 1 = A (AND with 1 leaves A unchanged), which is also true. Another pair is A . 0 = 0 and its dual A + 1 = 1, which is the answer to MCQ 2 of this chapter.

Long Questions

4. Compare and contrast half-adders and full-adders, including their truth tables, Boolean expressions, and circuit diagrams.

A half-adder adds two single bits, A and B, with no carry input. Its outputs are Sum = A XOR B and Carry = A AND B, giving the table: 0+0 = Sum 0, Carry 0; 0+1 = Sum 1, Carry 0; 1+0 = Sum 1, Carry 0; 1+1 = Sum 0, Carry 1. Its circuit is one XOR gate and one AND gate. A full-adder adds three bits, A, B and a carry-in Cin, so it handles the carry from a previous column. Its outputs are Sum = A XOR B XOR Cin and Carry-out = (A AND B) OR (Cin AND (A XOR B)). Its circuit is two XOR gates, two AND gates and one OR gate, and its table has 8 rows covering all combinations of A, B and Cin. In short, the half-adder is used only for the first column of an addition, while full-adders chain together for all higher columns.

Long Questions

5. How do Karnaugh maps simplify Boolean expressions? Provide a detailed example with steps.

Example: simplify F(A, B) = A' . B + A . B, where ' means NOT. Step 1: draw a 2-variable K-map with four cells for AB = 00, 01, 10, 11. Step 2: place 1s in the cells where F is 1: A'B is the cell 01, and AB is the cell 11, so the two 1s sit side by side in the B = 1 column. Step 3: group the two adjacent 1s into one group of two. Step 4: read the group: within it, A changes (0 to 1) so A drops out, while B stays 1, giving the single term B. The simplified function is therefore F = B, which needs no gates at all beyond a wire, instead of the original two AND gates

and one OR gate. The same grouping method works for larger maps, where groups of 4 or 8 drop two or three variables.

Long Questions

6. Design a 4-bit binary adder using both half-adders and full-adders. Explain each step with truth tables, Boolean expressions, and circuit diagrams.

To add two 4-bit numbers A3A2A1A0 and B3B2B1B0, chain one half-adder and three full-adders from the least significant bit upwards. Step 1 (bit 0): a half-adder adds A0 and B0, giving Sum S0 = A0 XOR B0 and carry C1 = A0 AND B0, since there is no incoming carry. Step 2 (bit 1): a full-adder adds A1, B1 and C1, giving S1 = A1 XOR B1 XOR C1 and carry C2 = (A1 AND B1) OR (C1 AND (A1 XOR B1)). Steps 3 and 4 repeat the full-adder for bits 2 and 3, producing S2, C3 and S3, C4, where C4 is the final overflow carry. The result is S3S2S1S0 with carry C4. Each stage's truth table is the half-adder table (4 rows) for bit 0 and the full-adder table (8 rows) for bits 1 to 3, and the circuit is the four blocks joined by the carry chain C1 to C4.

Long Questions

7. Simplify the following Boolean function using Boolean algebra rules: F(A, B) = A . B + A . B' (B' means NOT B, the overbar on B in the book)

Step 1: factor out A: F = A . (B + B'). Step 2: B + B' = 1, because a variable OR its complement is always 1 (complement law). Step 3: F = A . 1 = A (identity law). Answer: F = A.

Long Questions

8. Use De Morgan's laws to simplify the following function: F(A, B, C) = (A + B + A . C)' (the overbar covers the whole expression in the book)

Step 1: simplify inside first: A + A . C = A by the absorption law, so A + B + A . C

= A + B. Step 2: apply De Morgan's law to the complement of a sum: (A + B)' = A' . B'. Answer: F = A' . B' (NOT A AND NOT B).

Long Questions

9. Simplify the following expressions (overbars read from the book, page 68 render; ' means NOT): (a) (A + B)' . (A + B) (b) (A + B') . (A' + B) (c) A + A' . (B' + C) (d) A' . B + A . B (e) (A . B) + (A' . B')

(a) (A + B)' . (A + B): a value AND its complement is always 0, so the answer is 0. (b) Expand: (A + B') . (A' + B) = A . A' + A . B + B' . A' + B' . B = 0 + A . B + A' . B' + 0 = A . B + A' . B' (the XNOR function: 1 when A and B are equal). (c) A + A' . (B' + C): by A + A' . X = A + X, this equals A + B' + C. (d) A' . B + A . B = (A' + A) . B = 1 . B = B. (e) (A . B) + (A' . B') is already simplest: it is the XNOR function, 1 when A and B are equal, 0 otherwise.

Chapter 4: System Troubleshooting

Multiple Choice Questions (MCQs)

1. What is the first step in the systematic process of troubleshooting?

(a) Establish a Theory of Probable Cause

(b) Implement the Solution

(c) Identify Problem ✓

(d) Document Findings, Actions, and Outcomes

Correct answer: Identify Problem

Multiple Choice Questions (MCQs)

2. Why is effective troubleshooting important for maintaining systems?

(a) It helps save money on repairs

(b) It prevents the need for professional help

(c) It ensures systems operate smoothly and efficiently ✓

(d) It allows for more frequent system updates

Correct answer: It ensures systems operate smoothly and efficiently

Multiple Choice Questions (MCQs)

3. Which step involves coming up with a theory about what might be causing a problem?

(a) Test the Theory to Determine the Cause

(b) Establish a Theory of Probable Cause ✓

(c) Implement the Solution

(d) Verify Full System Functionality

Correct answer: Establish a Theory of Probable Cause

Multiple Choice Questions (MCQs)

4. After implementing a solution, what is the next step in the troubleshooting

process?

(a) Document Findings, Actions, and Outcomes

(b) Test the Theory to Determine the Cause

(c) Verify Full System Functionality

(d) Establish a Plan of Action to Resolve the Problem

Correct answer: Verify Full System Functionality

Multiple Choice Questions (MCQs)

5. Which of the following is an example of identifying a problem in troubleshooting?

(a) Testing a laptop battery by plugging in the power cord

(b) Coming up with a plan to replace a laptop battery

(c) Noticing that a laptop does not turn on when the power button is pressed

(d) Writing down that a laptop battery was replaced

Correct answer: Noticing that a laptop does not turn on when the power button is pressed

Multiple Choice Questions (MCQs)

6. Why is documenting findings, actions, and outcomes important in troubleshooting?

(a) It helps solve problems faster

(b) It provides a record for future reference

(c) It allows for more efficient testing

(d) It ensures the solution is implemented correctly

Correct answer: It provides a record for future reference

Multiple Choice Questions (MCQs)

7. What is the purpose of establishing a plan of action in troubleshooting?

(a) To identify the problem

(b) To verify full system functionality

(c) To determine the cause of the problem

(d) To decide on the steps needed to resolve the issue ✓

Correct answer: To decide on the steps needed to resolve the issue

Multiple Choice Questions (MCQs)

8. Why is troubleshooting important in computing systems?

(a) It ensures hardware components are always up to date

(b) It prevents the need for data backups

(c) It helps keep systems running smoothly and securely ✓

(d) It eliminates the need for software updates

Correct answer: It helps keep systems running smoothly and securely

Multiple Choice Questions (MCQs)

9. What does troubleshooting help prevent by quickly identifying and resolving issues?

(a) The need for professional help

(b) The need for software updates

(c) Downtime and lost productivity ✓

(d) The need for regular maintenance

Correct answer: Downtime and lost productivity

Multiple Choice Questions (MCQs)

10. Which of the following is an example of ensuring data integrity through troubleshooting?

(a) Identifying a software bug that causes incorrect database results ✓

(b) Replacing a faulty printer

(c) Using a cooling pad to prevent laptop overheating

(d) Updating the operating system regularly

Correct answer: Identifying a software bug that causes incorrect database results

Short Questions

1. What is the first step in the systematic process of troubleshooting, and why is it important?

The first step is to identify the problem. It is important because every later step depends on an accurate description of what is wrong: you must observe the symptoms, gather information from the user and the system, and define the problem clearly before you can form a theory about its cause. Without this step you may fix the wrong thing or miss the real fault, wasting time and possibly causing new problems.

Short Questions

2. After identifying a problem, what is the next step in troubleshooting, and how does it help in resolving the issue?

The next step is to establish a theory of probable cause: you use the symptoms and your knowledge to propose the most likely explanation for the fault. This helps because it narrows a wide field of possibilities down to a few testable ideas, so instead of trying random fixes you can test the most probable cause first and reach the solution systematically and quickly.

Short Questions

3. Describe the importance of testing a theory during the troubleshooting process. Provide an example.

Testing a theory is important because a theory is only a guess until it is checked: testing confirms or eliminates the suspected cause before you change anything

permanently, which avoids unnecessary replacements and wrong fixes. For example, if you theorize that a laptop will not turn on because its battery is dead, you test the theory by plugging in the power cord — if the laptop starts, the theory is confirmed and you know the battery is the problem; if it still does not start, you eliminate that theory and form a new one.

Short Questions

4. Explain what the 'Implement the Solution' step entails in troubleshooting.

Implementing the solution means carrying out the fix indicated by the confirmed theory, following an established plan of action. It entails deciding the exact steps needed to resolve the issue, applying them in order (for example replacing the faulty part, reinstalling the software, or changing the setting), and doing so carefully so the fix itself does not damage the system or lose data. Implementation comes only after the theory has been tested, not before.

Short Questions

5. Why is it necessary to verify full system functionality after implementing a solution?

Verification is necessary to confirm that the original problem is actually solved and that the fix has not created new problems elsewhere. You test the system fully — running the application, checking connected devices and confirming normal operation — rather than assuming success. Only after full functionality is verified do you document the findings, actions and outcomes, so the solution is proven and recorded for future reference.

Long Questions

1. Discuss the importance of troubleshooting in maintaining the smooth operation of systems, especially computing systems.

Troubleshooting is the systematic process of identifying, diagnosing and resolving problems, and it is what keeps systems operating smoothly and efficiently. In computing systems it matters for several reasons. First, it prevents downtime and lost productivity: quickly identifying and resolving issues means users and businesses are not left waiting while systems are down. Second, it maintains security: recognizing malware infections, applying operating system updates and fixing vulnerabilities keeps systems safe. Third, it ensures data integrity: finding, for example, a software bug that causes incorrect database results protects the accuracy of stored information. Fourth, it keeps systems running smoothly and securely without always needing costly professional help, since a trained user can fix many faults by following the systematic steps. In short, troubleshooting turns breakdowns from long, expensive disruptions into short, managed repairs.

Long Questions

2. Explain the systematic process of troubleshooting. Describe each step in detail.

The systematic process has six steps. (1) Identify the problem: observe symptoms, question the user, and define exactly what is wrong — for example, noticing that a laptop does not turn on when the power button is pressed. (2) Establish a theory of probable cause: propose the most likely explanation from the symptoms and your knowledge, such as a dead battery. (3) Test the theory to determine the cause: check the theory without yet making permanent changes, for example plugging in the power cord to see whether the laptop starts. (4) Establish a plan of action to resolve the problem and implement the solution: decide the steps needed and carry them out, for example replacing the battery. (5) Verify full system functionality: test the whole system to confirm the fault is gone and nothing else broke. (6) Document findings, actions and

outcomes: write down the problem, what was tried and what worked, providing a record for future reference. Following the steps in order avoids guesswork and repeated mistakes.

Long Questions

3. Using a case study where a printer is not printing, explain how you would identify the problem and establish a theory of probable cause.

Case study: a printer is not printing. Step 1 — Identify the problem: gather facts before guessing. Is the printer switched on and showing power? Is there paper and ink? Is it connected by cable or Wi-Fi, and does the computer show the printer as available? Does it print a test page, or does every job fail? Suppose the facts are: the printer has power and paper, but jobs sit in the queue and nothing prints, and the computer reports the printer as offline. Step 2 — Establish a theory of probable cause: the most likely explanations, in order, are (a) a connection problem between computer and printer (loose cable or dropped Wi-Fi), (b) a stalled print queue or wrong default printer selected, or (c) a missing or corrupt printer driver. The leading theory would be the connection problem, since the printer is reported offline. Step 3 (for completeness) would be to test it: check the cable or reconnect the Wi-Fi and send a test page. Only if the test confirms the theory would you implement the fix, verify full printing function, and document the outcome.

Long Questions

4. Discuss the importance of documenting findings, actions, and outcomes during the troubleshooting process.

Documentation matters because it provides a record for future reference. When the same problem recurs, the record shows what the symptoms were, which theories were tested, what the actual cause was and which fix worked, so the

fault can be solved faster the second time instead of repeating the whole investigation. It also helps others: colleagues, support staff and future technicians can learn from documented solutions and assist peers, and guides can be created from documented cases to share troubleshooting knowledge. Finally, documentation proves what was done — it records actions and outcomes for accountability, for example noting that a battery was replaced or a driver updated — which is essential in workplaces where systems must be audited and maintained over time.

Long Questions

5. Analyze the various ways troubleshooting is vital in computing systems, particularly in preventing downtime, ensuring data integrity, and improving security. Provide specific examples and scenarios to support your analysis.

Troubleshooting is vital in three connected ways. Preventing downtime: when a system fails, every minute of downtime costs productivity — for example, a frozen application in an office or a crashed server stops work for everyone depending on it, and systematic troubleshooting (identify, theorize, test, fix, verify) restores service far faster than random attempts. Ensuring data integrity: faults can silently corrupt information, and troubleshooting finds them — for example, identifying a software bug that causes incorrect database results protects every decision made from that data, while diagnosing a failing hard drive early allows data to be backed up before it is lost. Improving security: many faults are security threats, and troubleshooting recognizes them — for example, identifying and removing a malware infection, or applying operating system updates that close vulnerabilities, keeps the system secure. Together these make troubleshooting not just repair work but a core maintenance activity that keeps computing systems reliable, accurate and safe.

Long Questions

6. Describe basic software-related troubleshooting strategies, including handling application freezing and unresponsive peripherals.

Basic software troubleshooting starts with the simplest, least destructive steps. For an application that freezes: first wait briefly, since the program may recover; then close it through the task manager or by forcing it to quit, and reopen it; if freezing repeats, restart the computer, which clears temporary faults; then check for updates to the application and the operating system, and reinstall the application if the fault persists, since files may be corrupt. For unresponsive peripherals (for example a mouse, keyboard or printer that stops responding): check the physical connection (cable firmly attached, or wireless receiver and batteries), try a different port, restart the device and the computer, and check or reinstall the device driver, since drivers are the software that lets the operating system communicate with hardware. In both cases, work systematically: identify the symptom, form a theory (for example a driver fault), test it (try another port), implement the fix, verify full functionality, and document the outcome.

Long Questions

7. Explain how to recognize hardware failures, particularly focusing on RAM and hard drive issues.

Hardware failures are recognized from their symptoms, and RAM and hard drive faults show different patterns. RAM failure symptoms: frequent crashes or blue screens, the computer failing to boot or beeping at start-up, applications closing unexpectedly, and corrupted or garbled data on screen. To confirm, you test by reseating the memory modules, trying one module at a time, or running a memory diagnostic tool. Hard drive failure symptoms: strange noises such as clicking or grinding, very slow file access, files disappearing or becoming corrupt, frequent error messages when opening files, and the operating system failing to

start. To confirm, you check the drive's health with diagnostic tools, listen for abnormal sounds, and back up data immediately, since a failing drive can lose everything. In both cases the response follows the systematic process: identify the symptoms, theorize the faulty component, test it (swap or diagnose), replace the component if confirmed, verify the system works fully, and document the replacement.

Long Questions

8. Elaborate on the importance of maintaining software and addressing security threats.

Maintaining software means keeping it updated, clean and secure, and it is important for performance, reliability and safety. Operating system and application updates fix bugs and close security holes, so an un-updated system stays exposed to known attacks. Regular maintenance — removing unused programs, scanning for malware, and managing strong passwords — prevents the build-up of faults and threats that slow the system or compromise it. Addressing security threats means recognizing malware infections (unexpected pop-ups, sudden slowness, unknown programs), removing them with antivirus software, and applying operating system updates promptly. Together these practices keep systems running smoothly and securely, protect personal and organizational data, and prevent small issues from becoming serious breaches or breakdowns.

Long Questions

9. Describe common methods for identifying and removing malware infections and applying operating system updates for security.

To identify a malware infection, watch for its signs: sudden slowness, frequent pop-ups or crashes, unknown programs or browser toolbars, changed settings,

and unusual network activity. Confirm with a full scan using reputable, updated antivirus software, which can detect and quarantine suspicious files. To remove it: disconnect from the network if needed to stop the malware spreading, run the antivirus scan and let it quarantine or delete the infection, uninstall any unknown programs, and restart and rescan to verify the system is clean. To apply operating system updates for security: enable automatic updates where possible, or regularly check the system's update settings, download updates from the official source only (never from unknown links), install them, and restart the computer so the updates take effect; then verify the system runs normally. After both actions, document what was found and done, and keep backups so data can be recovered if an infection returns.

Long Questions

10. Describe the different data backup methods, including using external storage devices and cloud solutions.

Data backup means keeping copies of important data so it can be recovered after failure, theft or malware. External storage devices: copy data to a USB flash drive, an external hard drive or SSD, or other removable media. Advantages are that you control the copy, it works without the Internet, and recovery is fast; disadvantages are that the device can be lost, stolen or damaged in the same event (for example a fire) as the computer. Cloud solutions: store copies with an online backup service, which uploads data over the Internet and keeps it on remote servers. Advantages are that the copy is off-site and safe from local disasters, it is accessible from anywhere, and it can run automatically; disadvantages are dependence on an Internet connection and on the provider's security. A sound practice combines both: keep a local external copy for fast recovery and a cloud copy for protection against local loss, and test restores occasionally so the backup is known to work.

Chapter 5: Software System

Multiple Choice Questions (MCQs)

1. What is the primary function of an operating system?

(a) To create documents

(b) To manage hardware resources and provide a user interface ✓

(c) To perform calculations

(d) To design graphics

Correct answer: To manage hardware resources and provide a user interface

Multiple Choice Questions (MCQs)

2. Which software is used to enhance system performance and security?

(a) Operating system

(b) Utility software ✓

(c) Application software

(d) Device drivers

Correct answer: Utility software

Multiple Choice Questions (MCQs)

3. What role do device drivers play in a computer system?

(a) Manage files

(b) Facilitate communication between hardware devices and the operating system ✓

(c) Create presentations

(d) Enhance graphics performance

Correct answer: Facilitate communication between hardware devices and the operating system

Multiple Choice Questions (MCQs)

4. Which of the following is an example of application software?

(a) Microsoft Word

(b) BIOS

(c) Disk Cleanup

(d) Device Manager

Correct answer: Microsoft Word

Multiple Choice Questions (MCQs)

5. What is the main purpose of a spreadsheet software?

(a) To edit text documents

(b) To organize and analyze data

(c) To create visual content

(d) To enhance system security

Correct answer: To organize and analyze data

Multiple Choice Questions (MCQs)

6. How does utility software differ from application software?

(a) Utility software manages hardware, while application software performs specific tasks for users.

(b) Utility software creates documents, while application software manages hardware.

(c) Utility software performs specific tasks for users, while application software manages hardware.

(d) Utility software is free, while application software is paid.

Correct answer: Utility software manages hardware, while application software performs specific tasks for users.

Multiple Choice Questions (MCQs)

7. Which type of software would you use to design a logo?

(a) Operating system

(b) Spreadsheet software

(c) Graphic design software ✓

(d) Utility software

Correct answer: Graphic design software

Multiple Choice Questions (MCQs)

8. What is the function of system software?

(a) To facilitate communication between hardware and software ✓

(b) To perform specific tasks for the user

(c) To create visual content

(d) To organize and analyze data

Correct answer: To facilitate communication between hardware and software

Multiple Choice Questions (MCQs)

9. Why are operating system updates important?

(a) They increase screen brightness

(b) They add more fonts

(c) They enhance security and fix bugs ✓

(d) They improve battery life

Correct answer: They enhance security and fix bugs

Multiple Choice Questions (MCQs)

10. What is a common task you can perform using word processing software?

(a) Create and edit text documents ✓

(b) Manage hardware resources

(c) Enhance system performance

(d) Organize and analyze data

Correct answer: Create and edit text documents

Short Questions

1. Define system software and provide two examples.

System software is the basic software that enables a computer to run and manages its hardware and software resources, acting as the link between the user, the applications and the hardware. Two examples are an operating system (such as Windows or Linux), which manages hardware resources, provides the user interface and runs applications; and utility software (such as Disk Cleanup or antivirus software), which enhances system performance, security and maintenance.

Short Questions

2. Explain the primary functions of an operating system.

An operating system has three primary functions. It manages hardware resources: it controls the CPU, memory, storage and input/output devices, deciding which program uses them and when. It provides a user interface: the screens, windows, icons and menus through which the user gives commands and sees results. And it runs applications: it loads programs into memory, gives them the resources they need, and manages them while they execute, including managing files and folders the user creates.

Short Questions

3. What is utility software and why is it important?

Utility software is system software designed to enhance system performance, security and maintenance. It is important because computers slow down, accumulate unnecessary files and face security threats over time: utilities such

as disk cleanup remove junk files, disk defragmenters organize data for faster access, antivirus software detects and removes malware, and backup utilities protect data against loss. Without utilities, a system would gradually become slower, less secure and harder to maintain.

Short Questions

4. Describe the role of device drivers in a computer system.

A device driver is system software that facilitates communication between a hardware device and the operating system. The operating system cannot speak directly to every brand and model of printer, graphics card or mouse, so the driver translates the operating system's general commands into the specific instructions that device understands, and returns the device's results back to the operating system. Without the correct driver, the device may not work at all, or may work only partially.

Short Questions

5. Differentiate between system software and application software with examples.

System software runs and manages the computer itself, while application software performs specific tasks for the user. System software includes operating systems (Windows, Linux), utility software (Disk Cleanup, antivirus) and device drivers (printer driver, graphics driver); it works in the background and the user rarely interacts with it directly. Application software is what the user actually works with: word processing software (Microsoft Word) for documents, spreadsheet software for organizing and analyzing data, presentation software for slides, and graphic design software for images and logos. In short: system software makes the computer usable; application software makes it useful.

Short Questions

6. What are the main functions of spreadsheet software?

Spreadsheet software organizes and analyzes data in rows and columns of cells. Its main functions are storing data in structured tables; performing calculations automatically with formulas and functions (for example sums, averages and percentages) so results update when data changes; sorting and filtering data to find what is needed; and presenting data visually with charts and graphs that make patterns easy to see. It is used for budgets, marks records, stock lists and scientific data.

Short Questions

7. How can graphic design software be used in the field of education?

In education, graphic design software is used to create clear, attractive learning materials: teachers design posters, diagrams, worksheets and presentations that explain topics visually; students create project displays, charts and digital art that show their understanding; and schools design logos, banners and certificates. Visual material helps learners grasp ideas that are hard to explain in words alone, such as a labelled diagram of the heart or a chart of exam results, and it makes lessons more engaging and memorable.

Short Questions

8. What is the significance of data backups and how can they be performed?

Backups are significant because data can be lost at any time through hardware failure, malware, accidental deletion or theft, and without a copy that data is gone permanently. A backup is a second copy kept separately from the original, so the data can be restored after a loss. Backups can be performed by copying files to an external storage device such as a USB drive or external hard drive, or by using a cloud backup service that stores copies on remote servers over the

Internet; backup utility software can automate the process so it happens regularly without the user remembering each time.

Long Questions

1. Discuss the importance of system software in a computing system.

System software is important because it is what makes a computer work at all: without it, the hardware is a collection of parts that cannot cooperate, and application software has nothing to run on. The operating system manages hardware resources (CPU, memory, storage, devices), provides the user interface through which people operate the machine, and runs and coordinates applications. Utility software keeps the system performing well and secure by cleaning disks, removing malware and maintaining files. Device drivers let the operating system communicate with each specific hardware device. In short, system software facilitates communication between hardware and software, manages resources efficiently, and creates the stable platform on which every user task — writing, calculating, designing, browsing — depends.

Long Questions

2. Describe the roles of operating systems, utility software, and device drivers, providing examples of each.

The operating system is the core system software: it manages hardware resources (deciding which program gets CPU time and memory), provides the user interface (windows, icons, menus), runs applications, and manages files; examples are Windows, macOS and Linux. Utility software enhances system performance, security and maintenance: Disk Cleanup frees space by removing temporary files, antivirus software detects and removes malware, disk defragmenters organize data for faster access, and backup utilities protect data; each keeps the system healthy rather than doing a user task. Device drivers are

the translators between the operating system and hardware: a printer driver converts the operating system's print command into the exact instructions a particular printer model needs, a graphics driver lets the operating system use the graphics card's full capability, and a mouse driver interprets mouse movements; without the correct driver the device cannot communicate properly with the system.

Long Questions

3. Explain the differences between system software and application software.

System software and application software differ in purpose, user interaction and dependence. Purpose: system software manages and runs the computer (operating systems, utilities, drivers), while application software performs specific tasks for the user (word processing, spreadsheets, presentations, graphic design). Interaction: the user rarely interacts directly with system software, which works in the background, whereas application software is used directly for visible work such as writing a document. Dependence: application software cannot run without system software — a word processor needs an operating system to reach the CPU, memory and printer — while system software can run without any particular application. Examples make the split clear: Windows and Disk Cleanup are system software; Microsoft Word and a spreadsheet program are application software. Both are needed, but they serve different layers of the computer.

Long Questions

4. Describe the process of using utility software to optimize system performance and maintain security. Provide detailed steps and examples of common utility tools.

Using utility software follows a regular maintenance routine. Step 1: clean up the

disk with a tool such as Disk Cleanup, removing temporary files, recycle-bin contents and unused system files to free space. Step 2: check and repair the disk, scanning for file-system errors and bad sectors before they cause data loss. Step 3: defragment (on a traditional hard drive) so file fragments are reorganized and access becomes faster. Step 4: secure the system with antivirus software — update its definitions, run a full scan, and quarantine or remove any malware found. Step 5: manage start-up programs and uninstall unused applications, so the computer boots faster and fewer background programs consume resources. Step 6: back up important data with a backup utility to an external drive or cloud storage, and verify the backup can be restored. Step 7: keep the operating system and drivers updated, since updates fix bugs and security holes. Done regularly, this routine keeps the computer fast, stable and secure.

Long Questions

5. Explain how to install, update, and troubleshoot device drivers for hardware components.

Install: when new hardware is connected, the operating system often installs a driver automatically; otherwise insert the manufacturer's disc or download the driver from the manufacturer's official website, run the installer, and restart if asked. Always use the driver for the exact device model and operating system version. Update: open the device manager, find the device, and choose to update its driver, or download the newest version from the manufacturer; updates fix bugs, add features and improve compatibility, and should be installed especially when a device misbehaves after an operating system update. Troubleshoot: if a device stops working, first check the physical connection and power; then check the device manager for warning symbols, which indicate a driver problem; try updating the driver, rolling back to a

previous version if a new driver caused the fault, or uninstalling and reinstalling the driver completely. If the device still fails after a correct driver is installed, the fault is likely hardware, and the device may need repair or replacement.

Long Questions

6. Discuss the main functions of commonly used application software, such as word processing, spreadsheet, presentation, and graphic design applications.

Word processing software (for example Microsoft Word) creates and edits text documents: typing, formatting, spell-checking, inserting images and tables, and printing letters, reports and assignments. Spreadsheet software organizes and analyzes data in cells: entering values, calculating with formulas and functions, sorting and filtering, and showing data as charts for budgets, marks and stock records. Presentation software creates slide shows: combining text, images, charts and effects so a speaker can present ideas clearly to an audience, for lessons, business pitches and conferences. Graphic design software creates and edits visual content: drawing logos, posters and illustrations, editing photographs, and preparing layouts for print and screens, used by designers, advertisers and teachers. Together these four kinds cover the everyday work of a computer user: writing, calculating, presenting and designing.

Chapter 6: Introduction to Computer Networks

Multiple Choice Questions (MCQs)

1. What is the primary objective of computer networks?

(a) Increase computational power

(b) Enable resource sharing and data communication ✓

(c) Enhance graphic capabilities

(d) Improve software development

Correct answer: Enable resource sharing and data communication

Multiple Choice Questions (MCQs)

2. Which device is used to connect multiple networks and direct data packets between them?

(a) Switch

(b) Hub

(c) Router ✓

(d) Modem

Correct answer: Router

Multiple Choice Questions (MCQs)

3. Which layer of the OSI model is responsible for node-to-node data transfer and error detection?

(a) Physical Layer

(b) Data Link Layer ✓

(c) Network Layer

(d) Transport Layer

Correct answer: Data Link Layer

Multiple Choice Questions (MCQs)

4. What is the function of the Domain Name System (DNS)?

(a) Assign IP addresses dynamically

(b) Translate domain names to IP addresses ✓

(c) Secure data communication

(d) Monitor network traffic

Correct answer: Translate domain names to IP addresses

Multiple Choice Questions (MCQs)

5. Which method of data transmission uses a dedicated communication path?

(a) Packet Switching

(b) Circuit Switching ✓

(c) Full-Duplex

(d) Half-Duplex

Correct answer: Circuit Switching

Multiple Choice Questions (MCQs)

6. What is encapsulation in the context of network communication?

(a) Converting data into a secure format

(b) Wrapping data with protocol information ✓

(c) Monitoring network traffic

(d) Translating domain names to IP addresses

Correct answer: Wrapping data with protocol information

Multiple Choice Questions (MCQs)

7. Which protocol is used for reliable data transfer in the TCP/IP model?

(a) HTTP

(b) FTP

(c) TCP ✓

(d) UDP

Correct answer: TCP

Multiple Choice Questions (MCQs)

8. What is the main purpose of a firewall in network security?

(a) Convert data into a secure format

(b) Monitor and control network traffic ✓

(c) Assign IP addresses

(d) Translate domain names

Correct answer: Monitor and control network traffic

Multiple Choice Questions (MCQs)

9. Which network topology connects all devices to a central hub?

(a) Ring

(b) Mesh

(c) Bus

(d) Star ✓

Correct answer: Star

Multiple Choice Questions (MCQs)

10. What is a key benefit of using computer networks in businesses?

(a) Increase computational power

(b) Enable resource sharing and efficient communication ✓

(c) Enhance graphic capabilities

(d) Improve software development

Correct answer: Enable resource sharing and efficient communication

Short Questions

1. Define data communication and list its key components.

Data communication is the exchange of data between devices through a transmission medium, following agreed rules. Its key components are the sender (the device that sends the message), the receiver (the device that receives it), the message (the data itself), the transmission medium (the path, such as a cable or wireless link, over which the data travels), and the protocol (the set of rules governing the exchange). Without all five, reliable communication cannot take place.

Short Questions

2. Explain the role of routers in a computer network.

A router is a device that connects multiple networks and directs data packets between them. When data arrives, the router reads the destination address and chooses the best path across networks to forward it, so a message from a home network can reach a computer on the Internet. Routers therefore make communication between different networks possible, unlike switches, which only connect devices within one network.

Short Questions

3. What are the main functions of the Network Layer in the OSI model?

The Network Layer is responsible for delivering data across networks: it provides logical addressing (IP addresses) so devices on different networks can be identified, and it performs routing, choosing the path data packets take from source to destination. It sits between the Data Link Layer (node-to-node transfer on one link) and the Transport Layer (end-to-end delivery), and it is the layer where routers operate.

Short Questions

4. Describe the difference between packet switching and circuit switching.

In circuit switching, a dedicated communication path is set up between sender and receiver for the whole conversation and held until it ends, like a traditional telephone call: the path is reserved even during silences, which is reliable but wasteful. In packet switching, the message is broken into small packets that travel independently, possibly by different routes, and are reassembled at the destination, like Internet traffic: the network is used efficiently and can route around faults, but packets may arrive out of order or with delays.

Short Questions

5. What is the purpose of the Dynamic Host Configuration Protocol (DHCP)?

DHCP automatically assigns IP addresses and related network settings (such as subnet mask and gateway) to devices when they join a network. Its purpose is to remove manual configuration: without DHCP, every device's address would have to be set by hand, which is slow and error-prone, and two devices could accidentally receive the same address. With DHCP, a device simply requests an address and receives a valid, unique one automatically.

Short Questions

6. How does encapsulation ensure secure communication in a network?

Encapsulation wraps data with protocol information at each layer as it travels down the sender's network stack: each layer adds its own header (and sometimes trailer) containing addressing and control information, so the data becomes a packet that only the matching layer at the receiver can unwrap. This layered wrapping organizes the data, ensures it is delivered to the correct destination in the correct form, and supports secure communication because control information and checks travel with the data and the original message is protected inside the packet structure during transmission.

Short Questions

7. Differentiate between TCP and UDP in terms of data transfer reliability.

TCP (Transmission Control Protocol) provides reliable data transfer: it establishes a connection, numbers and acknowledges packets, retransmits lost ones, and delivers data in order and without errors, which suits file transfers, emails and web pages. UDP (User Datagram Protocol) provides fast but unreliable transfer: it sends packets without a connection, acknowledgements or retransmission, so packets may be lost or arrive out of order, which is acceptable for live video, voice calls and online games where speed matters more than perfect delivery.

Short Questions

8. Explain the importance of encryption in network security.

Encryption converts readable data into a coded form that only the intended recipient, holding the correct key, can decode. It is important because data travelling over a network can be intercepted: without encryption, anyone capturing the traffic could read passwords, messages and financial details, but with encryption the intercepted data is meaningless to the attacker. Encryption therefore protects confidentiality and is a core method of network security alongside firewalls and antivirus software.

Short Questions

9. What are the advantages of using a star topology in a network?

In a star topology all devices connect to a central hub or switch. Its advantages are easy fault detection and isolation: if one cable or device fails, only that device is affected and the rest of the network keeps working, and the faulty link is easy to find. It is also easy to add or remove devices without disrupting the network, and each device gets its own connection to the hub, so performance is consistent. Its main disadvantage is dependence on the central hub: if the hub

fails, the whole network fails.

Short Questions

10. How do firewalls contribute to network security?

A firewall monitors and controls network traffic passing between a trusted internal network and untrusted outside networks, such as the Internet. It applies security rules to incoming and outgoing packets, blocking unauthorized access attempts and suspicious traffic while allowing legitimate communication through. By acting as a barrier at the network boundary, it prevents attackers from reaching internal devices and stops malicious data from entering or leaving the network.

Long Questions

1. Discuss the objectives of computer networks and provide examples of how they facilitate resource sharing and data communication.

A computer network is a system of linked devices and computers that work together to facilitate communication. Its objectives are resource sharing, communication and data management. Resource sharing means multiple users share expensive resources instead of each needing their own: for example, several office computers share one printer and one Internet connection, and employees share files stored on a central server. Data communication means fast exchange of information between devices and users: for example, sending an email from one office to another, or students in different classrooms accessing the same learning material online. Together these objectives make work cheaper and faster: a business avoids buying a printer for every desk, a school gives every student access to the same digital library, and a hospital lets doctors in different departments view the same patient records.

Long Questions

2. In a Simplex communication system, assume data is transmitted at a rate of 500 bits per second (bps). Compute the time to transmit a message if: (a) it is of 10 kilobits. (b) it is of 10 kilobytes.

Time = amount of data divided by transmission rate. (a) 10 kilobits = 10,000 bits. Time = 10,000 / 500 = 20 seconds. (b) 10 kilobytes must first be converted to bits: 1 byte = 8 bits, so 10 kilobytes = 10 x 8 = 80 kilobits = 80,000 bits. Time = 80,000 / 500 = 160 seconds (2 minutes 40 seconds). The kilobyte message therefore takes eight times longer than the kilobit message, because a byte holds eight bits.

Long Questions

3. Describe how data is transmitted across computer networks using packet switching and circuit switching.

In circuit switching, before any data flows, a dedicated communication path is established between sender and receiver and reserved for the entire session, then released when the session ends. This guarantees a constant path and steady delivery, which suits voice calls, but it wastes capacity during pauses, since no one else can use the reserved path. In packet switching, the message is divided into small packets, each labelled with source and destination addresses; packets travel independently across the network, possibly taking different routes according to current traffic, and are reassembled in order at the destination. This uses the network efficiently, lets traffic route around failed links, and suits bursty data such as web browsing, but it needs extra work to reassemble packets and handle any that are lost or delayed. Modern computer networks, including the Internet, use packet switching.

Long Questions

4. Discuss the role and importance of protocols in data communication. Explain the functions of key protocols such as TCP/IP, HTTP, DNS, and DHCP.

Protocols are the agreed rules that govern data communication: without them, devices from different makers could not understand each other. They define message formats, addressing, error handling and the order of exchange, so communication is reliable and predictable. Key protocols: TCP/IP is the core protocol suite of the Internet — IP addresses packets and routes them across networks, while TCP provides reliable, ordered, error-checked delivery of the data. HTTP (HyperText Transfer Protocol) governs how web browsers request and receive web pages from servers. DNS (Domain Name System) translates human-readable domain names (such as a website name) into IP addresses computers use, so users need not remember numbers. DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network settings to devices joining a network, avoiding manual configuration and address clashes. Together these protocols let a user type a website name and reliably receive the correct page.

Long Questions

5. Evaluate different methods of network security, including firewalls, encryption, and antivirus software.

Firewalls monitor and control network traffic at the boundary between a trusted network and the outside world, blocking unauthorized access while allowing legitimate traffic; their strength is perimeter protection, but they cannot stop threats already inside the network or a user deliberately downloading malware. Encryption converts data into a code readable only with the correct key, protecting confidentiality even if traffic is intercepted; its strength is that stolen data stays useless, but it depends on keys being kept secret and adds processing overhead. Antivirus software detects, quarantines and removes

malware on individual devices, protecting against viruses that arrive by email, download or USB drive; its strength is device-level defence, but it must be kept updated to recognize new threats. No single method is enough: effective security combines all three — firewall at the boundary, encryption on the data, and antivirus on the devices — plus strong passwords and regular updates.

Long Questions

6. Describe real-world applications of computer networks in business, education, and healthcare.

In business, networks enable resource sharing and efficient communication: employees share printers, files and a single Internet connection, branches communicate by email and video conferencing, and online sales and banking run over networks. In education, networks give students and teachers access to shared learning resources: a school network lets every classroom reach the same digital library and learning platform, students submit work online, and classes can join video lessons with experts elsewhere. In healthcare, networks let hospitals share patient records between departments, doctors consult specialists remotely, and medical devices send monitoring data to central systems, so treatment is faster and better coordinated. In each field the same network benefits appear: shared resources, faster communication and access to information from many locations.

Long Questions

7. Compare and contrast the different types of network topologies (star, ring, bus, and mesh).

In a star topology all devices connect to a central hub or switch: a single device failure affects only that device and faults are easy to find, but the hub is a single point of failure and cabling costs are higher. In a ring topology each device

connects to two neighbours forming a closed loop, and data travels around the ring: it is simple and orderly, but one broken link or device can disrupt the whole ring. In a bus topology all devices share one central cable (the backbone): it is cheap and easy to install for small networks, but a break in the backbone disables the whole network and faults are hard to isolate. In a mesh topology every device connects to many or all others: it is the most reliable, since data can take alternative routes if a link fails, but it is the most expensive and complex to cable and manage. Choice depends on need: star for most offices, bus for small cheap setups, ring where order matters, and mesh where reliability is critical.

Long Questions

8. Consider a shift cipher with a shift amount of 4. (a) Encrypt the message 'SECURITY'. (b) Decrypt the message 'WMXYVMI'.

(a) Encrypt by shifting each letter 4 places forward in the alphabet: S->W, E->I, C->G, U->Y, R->V, I->M, T->X, Y->C. The encrypted message is WIGYVMXC. (b) Decrypt by shifting each letter 4 places back: W->S, M->I, X->T, Y->U, V->R, M->I, I->E. The decrypted result is SITURIE. Note: SITURIE is not an English word, and the book's ciphertext WMXYVMI does not decrypt to a recognizable word under a shift of 4; the ciphertext as printed appears to contain a misprint, since encrypting SECURITY with a shift of 4 gives WIGYVMXC, not WMXYVMI. The mechanical decryption above follows the book's stated shift exactly.

Long Questions

9. An IPv4 address is a 32-bit number. Calculate the total number of unique IPv4 addresses possible. (a) Show the calculation for the total number of IPv4 addresses. (b) How many addresses are left if 10% of the total addresses are reserved for special purposes?

(a) Each of the 32 bits can be 0 or 1, so the total number of unique addresses is 2 to the power 32. 2^32 = 4,294,967,296 — approximately 4.3 billion addresses.

(b) 10% reserved for special purposes = 10% of 4,294,967,296 = 429,496,729.6, about 429,496,730 addresses. Addresses left = 4,294,967,296 - 429,496,730 = 3,865,470,566, about 3.87 billion addresses (90% of the total). This calculation shows why the 32-bit IPv4 space, once thought ample, is now exhausted in practice, which is why the much larger IPv6 space was introduced.

Chapter 7: Computational Thinking

Multiple Choice Questions

1. Which of the following best defines computational thinking?

(a) A method of solving problems using mathematical calculations only.

(b) A problem-solving approach that employs systematic, algorithmic, and logical thinking. ✓

(c) A technique used exclusively in computer programming.

(d) An approach that ignores real-world applications.

Correct answer: A problem-solving approach that employs systematic, algorithmic, and logical thinking.

Multiple Choice Questions

2. Why is problem decomposition important in computational thinking?

(a) It simplifies problems by breaking them down into smaller, more manageable parts. ✓

(b) It complicates problems by adding more details.

(c) It eliminates the need for solving the problem.

(d) It is only useful for simple problems.

Correct answer: It simplifies problems by breaking them down into smaller, more manageable parts.

Multiple Choice Questions

3. Pattern recognition involves:

(a) Finding and using similarities within problems ✓

(b) Ignoring repetitive elements

(c) Breaking problems into smaller pieces

(d) Writing detailed algorithms

Correct answer: Finding and using similarities within problems

Multiple Choice Questions

4. Which term refers to the process of ignoring the details to focus on the main idea?

(a) Decomposition

(b) Pattern recognition

(c) Abstraction ✓

(d) Algorithm design

Correct answer: Abstraction

Multiple Choice Questions

5. Which of the following is a principle of computational thinking?

(a) Ignoring problem understanding

(b) Problem simplification ✓

(c) Avoiding solution design

(d) Implementing random solutions

Correct answer: Problem simplification

Multiple Choice Questions

6. Algorithms are:

(a) Lists of data

(b) Graphical representations

(c) Step-by-step instructions for solving a problem ✓

(d) Repetitive patterns

Correct answer: Step-by-step instructions for solving a problem

Multiple Choice Questions

7. Which of the following is the first step in problem-solving according to computational thinking?

(a) Writing the solution

(b) Understanding the problem ✓

(c) Designing a flowchart

(d) Selecting a solution

Correct answer: Understanding the problem

Multiple Choice Questions

8. Flowcharts are used to:

(a) Code a program

(b) Represent algorithms graphically ✓

(c) Solve mathematical equations

(d) Identify patterns

Correct answer: Represent algorithms graphically

Multiple Choice Questions

9. Pseudocode is:

(a) A type of flowchart

(b) A high-level description of an algorithm using plain language ✓

(c) A programming language

(d) A debugging tool

Correct answer: A high-level description of an algorithm using plain language

Multiple Choice Questions

10. Dry running a flowchart involves:

(a) Writing the code in a programming language

(b) Testing the flowchart with sample data ✓

(c) Converting the flowchart into pseudocode

(d) Ignoring the flowchart details

Correct answer: Testing the flowchart with sample data

Short Questions

1. Define computational thinking.

Computational thinking is a problem-solving approach that employs systematic, algorithmic and logical thinking. It means tackling a problem the way a computer scientist would: understanding it clearly, breaking it into parts, spotting patterns, ignoring irrelevant detail, and designing a clear step-by-step solution that could be carried out by a person or a computer.

Short Questions

2. What is decomposition in computational thinking?

Decomposition is breaking a large, complex problem into smaller, more manageable parts, each of which can be understood and solved separately. For example, organizing a school event decomposes into booking a venue, arranging transport, preparing food, and inviting guests. It simplifies problems because each small part is easier to solve than the whole, and the solutions combine into the full answer.

Short Questions

3. Explain pattern recognition with an example.

Pattern recognition is finding and using similarities within problems: when you notice that a new problem resembles one you have solved before, you can reuse the earlier solution pattern. For example, once you learn the pattern for finding the largest of three numbers (compare the first two, keep the larger, compare it with the third), you can apply the same pattern to find the largest score in a class list, or the oldest student in a group — the data changes, but the pattern stays the same.

Short Questions

4. Describe abstraction and its importance in problem-solving.

Abstraction is ignoring the details to focus on the main idea: keeping only the information relevant to the problem and hiding the rest. It is important because real problems carry far more detail than a solution needs — for example, a map for navigation shows roads and landmarks but hides trees, house colours and shop interiors, and a driver can still reach the destination. By abstracting, you simplify the problem and avoid being distracted by irrelevant detail.

Short Questions

5. What is an algorithm?

An algorithm is a step-by-step set of instructions for solving a problem, written clearly enough that it can be followed exactly and always gives the same result for the same input. Examples are a recipe for cooking, directions for reaching a place, and the steps for adding two numbers. An algorithm must be unambiguous, finite (it must end), and effective (each step must be doable).

Short Questions

6. How does problem understanding help in computational thinking?

Problem understanding is the first step of problem-solving, and it helps because you cannot design a correct solution for a problem you have not understood: you must know what is given (the inputs), what is required (the output), and what constraints apply. For example, before writing a grading algorithm you must know the exact mark ranges for each grade; without that, even a perfect algorithm would give wrong grades. Understanding the problem also prevents wasted work on the wrong solution.

Short Questions

7. What are flowcharts and how are they used?

A flowchart is a graphical representation of an algorithm, using standard shapes connected by arrows: ovals for start and end, parallelograms for input and output, rectangles for processing steps, and diamonds for decisions. Flowcharts are used to design, visualize and communicate an algorithm before coding, because the flow of steps and decisions can be seen at a glance, errors are easier to spot, and the design can be discussed with others who may not read code.

Short Questions

8. Explain the purpose of pseudocode.

Pseudocode is a high-level description of an algorithm using plain language, written in a structured, code-like style but without the strict syntax of a programming language. Its purpose is to plan an algorithm in a form that is easy to read, check and translate into any programming language later. It bridges the gap between a human idea and actual code: you work out the logic first, then implement it in Python, Java or another language.

Short Questions

9. How do you differentiate between flowcharts and pseudocode?

Both describe algorithms, but flowcharts represent them graphically with shapes and arrows, while pseudocode describes them in written, plain-language steps. Flowcharts are better for seeing the overall flow and decision branches at a glance, and for explaining an algorithm to others visually; pseudocode is better for detailed, step-heavy logic and converts more directly into code. In practice they complement each other: a flowchart shows the structure, pseudocode spells out the details.

Short Questions

10. What is a dry run and why is it important?

A dry run is manually going through an algorithm or flowchart with sample data, step by step, tracking how values change, to identify any errors before the program is written or run. It is important because it catches logic errors early — wrong conditions, missed cases, incorrect calculations — when they are still cheap and easy to fix, and it confirms that the algorithm actually produces the expected output for known inputs.

Short Questions

11. Describe LARP and its significance in learning algorithms.

LARP stands for Logic of Algorithm for Resolution of Problems. It is a fun, interactive way to learn how algorithms work by actually running them and seeing the results — learners act out or simulate the steps of an algorithm, for example by physically sorting cards to follow a sorting algorithm, instead of only reading about it. Its significance is that it makes abstract algorithmic thinking concrete and memorable: by experiencing each step, learners understand why the algorithm works and where it can go wrong.

Short Questions

12. List and explain two debugging techniques.

Two debugging techniques are dry running and simulation. Dry running means manually tracing the algorithm with sample data, recording each variable's value at each step, to find where the actual behaviour diverges from the expected behaviour. Simulation means using a computer program or model to imitate the real process — for example, running a flowchart in a simulator or testing code with test data — so errors can be observed safely without affecting a real system. Both techniques find and help fix errors (bugs) in an algorithm or

flowchart; a third approach is testing with boundary values, such as the smallest and largest allowed inputs.

Long Questions

1. Write an algorithm to assign a grade based on the marks obtained by a student. The grading system follows these criteria: 90 and above: A+; 80 to 89: A; 70 to 79: B; 60 to 69: C; Below 60: F.

Step 1: Start. Step 2: Input the student's marks. Step 3: If marks are 90 or above, assign grade A+. Step 4: Else, if marks are between 80 and 89, assign grade A. Step 5: Else, if marks are between 70 and 79, assign grade B. Step 6: Else, if marks are between 60 and 69, assign grade C. Step 7: Else (marks below 60), assign grade F. Step 8: Output the grade. Step 9: Stop. Dry run check: marks 85 reach Step 4 (80 to 89) and give A; marks 59 fall through all conditions to Step 7 and give F; marks 90 give A+ at Step 3.

Long Questions

2. Explain how you would use algorithm design methods, such as flowcharts and pseudocode, to solve a complex computational problem. Illustrate your explanation with a detailed example.

To solve a complex problem, first understand it and decompose it into smaller parts, then design the solution with flowcharts and pseudocode before coding. Example: a library wants to find overdue books. Decomposition: (a) read each book's due date and today's date, (b) compare them, (c) list books past their due date, (d) print borrower details. Flowchart: start (oval) -> input today's date and book records (parallelogram) -> for each book, is due date before today? (diamond) -> if yes, add to overdue list (rectangle); if no, skip -> more books? (diamond) -> if yes, repeat; if no, print overdue list (parallelogram) -> stop (oval). Pseudocode: Start; Input today; For each book record: If due date < today

Then add book to overdue list; End For; Print overdue list; Stop. The flowchart shows the flow and decisions visually, the pseudocode spells out the exact logic, and a dry run with sample dates verifies the design before any code is written.

Long Questions

3. Define computational thinking and explain its significance in modern problem-solving. Provide examples to illustrate how computational thinking can be applied in different fields.

Computational thinking is a problem-solving approach that employs systematic, algorithmic and logical thinking, built on four pillars: decomposition (breaking problems into parts), pattern recognition (finding similarities), abstraction (focusing on relevant detail only) and algorithm design (creating step-by-step solutions). Its significance is that it gives a reliable method for tackling complex problems in any field, not just computing: it turns vague, overwhelming problems into clear, solvable steps. In business, a shopkeeper decomposes monthly sales analysis into collecting daily sales, grouping by product, finding patterns (which products sell best on weekends) and designing a reordering routine. In medicine, doctors use pattern recognition (symptoms matching known conditions) and abstraction (ignoring irrelevant patient details) to diagnose. In science, researchers design algorithms to process experimental data. In everyday life, planning a journey uses decomposition (route, transport, timing), abstraction (ignoring irrelevant streets) and an algorithm (the sequence of turns). Computational thinking is therefore a general life skill for the modern world, where data and systems surround every decision.

Long Questions

4. Discuss the concept of decomposition in computational thinking. Why is it important?

Decomposition is the process of breaking a large, complex problem into smaller, more manageable sub-problems that can each be solved separately and then combined. It is important for three reasons. First, it simplifies: a small part is easier to understand, solve and test than the whole — for example, building a school website decomposes into designing pages, writing content, adding images and testing links, and each task can be given to a different person. Second, it enables reuse: a sub-problem solved once (such as a login routine) can be reused wherever it appears. Third, it makes errors easier to find: when a part fails, you know which part to fix, rather than searching the whole problem. Without decomposition, complex problems remain overwhelming and error-prone; with it, even very large systems become a collection of solvable pieces.

Long Questions

5. Explain pattern recognition in the context of computational thinking. How does identifying patterns help in problem-solving?

Pattern recognition is finding and using similarities within problems — noticing that parts of a new problem match parts of problems already solved, or that data contains repeating trends. It helps in three ways. First, it enables reuse: if you recognize that computing a class average follows the same pattern as computing a shop's daily average (sum the values, divide by the count), you do not need to invent a new method. Second, it reveals predictions: spotting that sales rise every weekend lets a shopkeeper stock accordingly. Third, it simplifies algorithms: a sorting pattern learned once (compare neighbours, swap if out of order) applies to numbers, names and dates alike. In short, patterns turn new problems into familiar ones, so solutions become faster, more reliable and easier to generalize.

Long Questions

6. What is an abstraction in computational thinking? Discuss its importance and provide examples of how abstraction can be used to simplify complex problems.

Abstraction is the process of ignoring details to focus on the main idea — keeping only the information relevant to the problem and hiding the rest. Its importance is that real problems carry overwhelming detail, and abstraction lets you reason about the essentials: without it, every problem would be too complex to model or solve. Examples: a road map abstracts a city to roads and landmarks, hiding trees and building colours, yet it is exactly what a driver needs; a weather app abstracts atmospheric physics into a simple forecast (sunny, 28 degrees), hiding the equations behind it; in programming, a 'print' command abstracts away the printer's hardware details, letting the programmer focus on what to print. In each case abstraction simplifies the problem to a manageable model while preserving what matters for the solution.

Long Questions

7. Describe what an algorithm is and explain its role in computational thinking. Provide a detailed example of an algorithm for solving a specific problem, and draw the corresponding flowchart.

An algorithm is a step-by-step set of instructions for solving a problem, precise enough to be followed exactly and to finish with a correct result. Its role in computational thinking is central: decomposition, pattern recognition and abstraction prepare the problem, but the algorithm is the actual solution — the ordered procedure that a person or computer executes. Example: finding the largest of three numbers. Algorithm: Step 1: Start. Step 2: Input A, B, C. Step 3: Set Largest = A. Step 4: If B > Largest, set Largest = B. Step 5: If C > Largest, set Largest = C. Step 6: Output Largest. Step 7: Stop. Corresponding flowchart: Start (oval) -> Input A, B, C (parallelogram) -> Largest = A (rectangle) -> Is B >

Largest? (diamond) -> if yes, Largest = B (rectangle), then -> Is C > Largest? (diamond) -> if yes, Largest = C (rectangle), then -> Output Largest (parallelogram) -> Stop (oval). A dry run with A=5, B=9, C=7 gives Largest=9, confirming the design.

Long Questions

8. Compare and contrast flowcharts and pseudocode as methods for algorithm design. Discuss the advantages and disadvantages of each method, and provide examples where one might be preferred over the other.

Flowcharts represent algorithms graphically with shapes (ovals, parallelograms, rectangles, diamonds) joined by arrows; pseudocode describes them in plain-language, structured written steps. Flowchart advantages: the flow and decision branches are visible at a glance, errors in logic are easy to spot, and non-programmers can follow them — for example, explaining a school admission process to office staff. Disadvantages: large algorithms make huge, cluttered diagrams that are hard to draw, modify and print. Pseudocode advantages: it is compact, easy to edit, handles detailed logic well, and translates almost directly into code — for example, drafting a sorting routine before coding it in Python. Disadvantages: it has no strict standard, so different writers use different styles, and it cannot show overall flow as instantly as a diagram. In practice, use a flowchart when the audience needs the big picture or the algorithm is decision-heavy, and pseudocode when the logic is detailed or coding is the next step; many designers use both.

Long Questions

9. Explain the concept of a dry run in the context of both flowcharts and pseudocode. How does performing a dry run help in validating the correctness of an algorithm?

A dry run is manually going through an algorithm with sample data, tracking every variable step by step, without running it on a computer. With a flowchart, you follow the arrows with a chosen input, writing down the value of each variable at each shape, and checking that the path taken matches expectations. With pseudocode, you read each written step in order, updating a trace table of variable values as you go. It validates correctness in three ways: it confirms the algorithm produces the expected output for known inputs; it reveals logic errors (wrong conditions, missing cases, infinite loops) before coding, when fixes are cheap; and it tests edge cases, such as the smallest, largest and boundary inputs (for example marks of 59, 60, 89 and 90 in a grading algorithm). An algorithm that passes dry runs on varied sample data is far more likely to be correct when finally coded.

Long Questions

10. What is LARP? Discuss its importance in learning and practicing algorithms.

LARP stands for Logic of Algorithm for Resolution of Problems: a fun, interactive way to learn algorithms by actually running them and seeing the results, rather than only reading about them. Learners might physically act out an algorithm — for example, students standing in a line and swapping places to perform a sorting algorithm, or moving tokens along a flowchart drawn on the floor. Its importance is that it makes abstract logic concrete: learners feel why each step happens, see where an algorithm goes wrong, and remember the process far better than from a textbook description. It also builds confidence, because learners experience success in executing an algorithm themselves before they are asked to write code for it, and it encourages teamwork and discussion about alternative solutions.

Long Questions

11. How does LARP enhance the understanding and application of computational thinking principles? Provide a scenario where LARP can be used to improve an algorithm.

LARP enhances computational thinking because acting out an algorithm forces learners to use all four pillars at once: they decompose the task into physical steps, recognize patterns in repeated actions, abstract away irrelevant detail (focusing on the rule, not the props), and follow a precise algorithm. Scenario: a class learning to sort exam marks. Students each hold a number card and physically perform a bubble-sort: compare neighbours, swap if out of order, repeat passes until sorted. While acting it out, they notice a pattern — after each pass, the largest remaining number has 'bubbled' to its final place — and they improve the algorithm by stopping each next pass one position earlier, since that part is already sorted. This optimization, discovered through physical experience, is then written back into the pseudocode and flowchart. LARP thus turns abstract principles into lived experience and produces genuinely better algorithms.

Chapter 8: Web Development with HTML, CSS, and JavaScript

Multiple Choice Questions

1. Which of the following tag is not a correct HTML tag? (a) (b) (c) <head> (d) <footer> — NOTE on the question as printed

As printed, this question has no correct option, because all four options are correct HTML tags: is a block container, is an inline container, <head> holds a page's metadata (title, links, scripts), and <footer> marks a page or section footer. None of them is incorrect. A student answering this question should note the error in the book rather than guess: if the intended distractor was a made-up tag, it is not present among the options given.

Multiple Choice Questions

2. What does CSS stand for?

(a) Cascading Style Sheets 

(b) Computer Style Sheets

(c) Creative Style Sheets

(d) Colorful Style Sheets

Correct answer: Cascading Style Sheets

Multiple Choice Questions

3. Which of the following tag is used to create a hyperlink in HTML?

(a) <link>

(b) <a> 

(c) <href>

(d) <nav>

Correct answer: <a>

Multiple Choice Questions

4. Which property is used to change the background color in CSS?

(a) color

(b) background-color ✓

(c) bgcolor

(d) background

Correct answer: background-color

Multiple Choice Questions

5. Which HTML attribute is used to define inline styles?

(a) class

(b) style ✓

(c) font

(d) styles

Correct answer: style

Multiple Choice Questions

6. Which of the following is the correct syntax for a CSS rule?

(a) selector {property: value;} ✓

(b) selector: {property=value;}

(c) selector {property=value}

(d) selector: {property: value;}

Correct answer: selector {property: value;}

Multiple Choice Questions

7. In JavaScript, which markup is used for comments? (a) /* */ (b) // (c) <-- (d) /* */ — NOTE on the question as printed

As printed, this question has no single correct option: options (a) and (d) are

identical (/* */), and both /* */ (multi-line comments) and // (single-line comments) are valid JavaScript comment markers. Option (c) <-- is not a JavaScript comment (it resembles an HTML comment opening). So the valid JavaScript comment forms among the options are /* */ and //; the question's duplicated option makes a single intended answer impossible, and a student should note this rather than treat one of them as uniquely correct.

Multiple Choice Questions

8. How do you include JavaScript in an HTML document?

(a) <script src="script.js"></script> ✓

(b) <java src="script.js"></java>

(c) <js src="script.js"></js>

(d) <code src="script.js"></code>

Correct answer: <script src="script.js"></script>

Multiple Choice Questions

9. Which HTML tag is used to create an unordered list?

(a)

(b) ✓

(c)

(d) <list>

Correct answer:

Multiple Choice Questions

10. Which tag is used to display a horizontal line in HTML?

(a)

(b) <hr> ✓

(c) <line>

(d) <hline>

Correct answer: <hr>

Short Questions

1. What is the purpose of the <head> tag in HTML?

The <head> tag holds metadata about the web page — information that is not displayed as page content but that the browser needs. It contains the page title (shown on the browser tab), links to external CSS files, internal styles, scripts, and character-set declarations. Everything the visitor reads appears in the <body>, while the <head> prepares the page behind the scenes.

Short Questions

2. Explain the difference between an ordered list and an unordered list in HTML.

An ordered list, created with <ol>, shows items in a numbered sequence (1, 2, 3...), used when order matters, such as steps in a recipe. An unordered list, created with <ul>, shows items with bullet points instead of numbers, used when order does not matter, such as a shopping list. In both cases each item is written inside an <li> tag.

Short Questions

3. How do you add a comment in CSS?

A CSS comment is written between /* and */ markers, for example: /* This styles the main heading */. Comments are ignored by the browser, so they are used to explain the stylesheet to other developers or to temporarily disable a rule while testing, without deleting it.

Short Questions

4. What are the different ways to apply CSS to an HTML document?

There are three ways. Inline CSS applies a style directly to one element using the style attribute, for example <p style='color: blue;'>. Internal (embedded) CSS places rules inside a <style> element in the document's <head>, applying to that page only. External CSS places rules in a separate .css file linked from the <head> with <link>, so many pages can share one stylesheet; this is the preferred method for real websites because one file change updates every page.

Short Questions

5. How can you include JavaScript in an HTML file?

JavaScript can be included in two ways. Internal JavaScript is written inside a <script> element in the HTML document, usually before the closing </body> tag. External JavaScript is stored in a separate .js file and linked with <script src='script.js'></script>. The external method is preferred for larger sites, since the same script can be reused on many pages and kept separate from the page content.

Short Questions

6. Describe the syntax for creating a hyperlink in HTML.

A hyperlink is created with the anchor tag <a> and its href attribute, which holds the destination address: Visit Example. The text between the opening and closing tags is the clickable link text. The href can point to another website, another page on the same site, or a section within the same page (using an id, such as href='#section1').

Short Questions

7. What is the function of the tag in HTML?

The tag is a block-level container that groups other HTML elements together so they can be styled or positioned as one section — for example,

grouping a heading, a paragraph and an image into a 'header' or 'article' block. It has no meaning or appearance by itself; its purpose is structural, letting CSS target the whole group at once.

Short Questions

8. How do you link an external CSS file to an HTML document?

An external CSS file is linked from the document's <head> using the <link> element: <link rel='stylesheet' href='styles.css'>. The rel attribute says the linked file is a stylesheet, and href gives the file's path. Once linked, every CSS rule in that file applies to the page, and the same file can be linked from many pages.

Short Questions

9. What is the use of the <table> tag in HTML?

The <table> tag creates a table for displaying data in rows and columns. Inside it, <tr> defines each row, <th> defines header cells (bold, centred headings), and <td> defines ordinary data cells. It is used for tabular data such as a marks sheet, a timetable or a price list, and can be styled with CSS for borders, colours and spacing.

Short Questions

10. Explain the box model in CSS.

The CSS box model describes how every HTML element is treated as a rectangular box made of four layers, from inside out: the content (the actual text or image), the padding (space between the content and the border), the border (the edge around the padding), and the margin (space outside the border separating the element from its neighbours). The total space an element occupies is the sum of all four layers, so setting width, padding, border and

margin together determines the element's real size and spacing on the page.

Long Questions

1. Discuss the fundamental differences between HTML, CSS, and JavaScript in the context of web development.

HTML, CSS and JavaScript are the three core technologies of web development, each with a distinct role. HTML (HyperText Markup Language) is the structure: it defines what content exists on a page — headings from <h1> to <h6>, paragraphs, lists, links, images and tables — like the walls and rooms of a house. CSS (Cascading Style Sheets) is the presentation: it controls how that content looks — colours, fonts, sizes, layouts, spacing and the box model — like paint, decoration and furnishing. JavaScript is the behaviour: it is a programming language that makes pages interactive and engaging, responding to clicks, checking forms, changing content without reloading, and handling events. A page can exist with HTML alone, looks plain without CSS, and stays static without JavaScript; together they produce a complete, attractive, interactive website. Front-end development is precisely this combination, while back-end development manages the behind-the-scenes servers, databases and application logic.

Long Questions

2. Explain the process of setting up a development environment for web development. By discussing the necessary softwares and tools.

Setting up a web development environment needs only a few tools. (1) A text editor or code editor, such as Visual Studio Code, for writing HTML, CSS and JavaScript with helpful features like colour-coding and auto-completion. (2) A modern web browser, such as Chrome or Firefox, for viewing and testing pages, including its developer tools for inspecting elements and debugging. (3) A folder

structure on the computer: one project folder holding index.html, a styles.css file and a script.js file, so the three technologies stay organized and can be linked together. (4) Optionally, a local server or live-preview extension for testing pages that load external resources, and version-control software such as Git for tracking changes. The process is: install the editor and browser, create the project folder and the three starter files, link the CSS with <link> and the JavaScript with <script src>, open index.html in the browser, and then write, save and refresh repeatedly to see each change.

Long Questions

3. Create a basic HTML page that includes a header, a paragraph, an image, and a hyperlink.

A basic page is written as follows. <!DOCTYPE html> declares the document type. <html> wraps the whole page. Inside <head>, <title>My First Page</title> sets the browser-tab title. Inside <body>: a header, <h1>Welcome to My Website</h1>; a paragraph, This is my first web page, created with HTML. ; an image, <img src='photo.jpg' alt='A photo'>, where src gives the image file and alt gives text if the image cannot load; and a hyperlink, Visit Example. Saved as index.html and opened in a browser, this displays a heading, a paragraph, the image and a clickable link, demonstrating the four building blocks of a page.

Long Questions

4. How do you style a table using CSS? Create a sample table and apply styles to it.

A table is created in HTML with <table>, rows with <tr>, headers with <th> and data cells with <td>, for example a marks table with Name and Marks columns. It is styled in CSS by targeting the table elements: table { border-collapse:

collapse; width: 100%; } merges the cell borders into single lines and stretches the table; th, td { border: 1px solid black; padding: 8px; text-align: left; } gives every cell a border and breathing space; th { background-color: darkblue; color: white; } highlights the header row; and tr:nth-child(even) { background-color: lightgrey; } shades alternate rows so the table is easy to read. The result is a clean, readable table rather than the browser's plain default.

Long Questions

5. Describe the different CSS selectors and provide examples of each.

CSS selectors choose which elements a rule applies to. The element (type) selector targets all elements of a type: p { color: blue; } colours every paragraph. The class selector targets elements with a given class attribute, using a dot: .highlight { background-color: yellow; } styles every element whose class is 'highlight', and one class can be reused on many elements. The id selector targets the single element with a given id, using a hash: #main-title { font-size: 32px; } styles only the element with id 'main-title'. The universal selector * targets every element: * { margin: 0; } resets margins everywhere. The group selector applies one rule to several selectors at once: h1, h2, h3 { font-family: Arial; }. The descendant selector targets elements inside another: div p { color: green; } styles only paragraphs inside a . Choosing the right selector keeps styles precise and reusable.

Long Questions

6. Explain the process of creating a responsive web page using CSS with the help of examples and explanations.

A responsive web page adapts its layout to the visitor's screen size — phone, tablet or desktop — instead of forcing one fixed layout. The process has four parts. (1) Use the viewport meta tag in the <head> so mobile browsers size the

page to the device width. (2) Use flexible layouts instead of fixed pixel widths: percentages and flexible boxes (display: flex) let columns shrink and grow, for example a two-column layout that becomes one column on a narrow screen. (3) Use flexible images: img { max-width: 100%; height: auto; } stops images overflowing small screens. (4) Use media queries to apply different rules at different widths, for example: @media (max-width: 600px) { .column { width: 100%; } } stacks columns vertically on phones while they sit side by side on desktops. Tested by resizing the browser, the page then stays readable and usable on every device.

Long Questions

7. Write a JavaScript function that changes the background color of a web page when a button is clicked. Provide the complete code and explain how it works.

HTML: <button onclick='changeColor()'>Change Background</button> creates a button that calls the function changeColor when clicked. JavaScript: function changeColor() { document.body.style.backgroundColor = 'lightblue'; } — document.body refers to the page's body element, .style.backgroundColor sets its CSS background-color property, and assigning 'lightblue' changes the colour. How it works: the browser loads the page with its default background; when the user clicks the button, the click event triggers the onclick handler, which runs changeColor, which immediately changes the body's background to light blue without reloading the page. The function can be extended, for example by choosing a random colour from a list each time it is clicked.

Long Questions

8. How do you add animations and transitions using CSS? Provide examples and explain the properties involved.

CSS transitions make a property change smoothly over time instead of instantly:

for example, button { background-color: blue; transition: background-color 0.5s; } makes the button's colour fade to a new value over half a second when a hover rule changes it (button:hover { background-color: red; }). The transition property names the property (or 'all'), the duration, and optionally a timing function and delay. CSS animations are more powerful: they use @keyframes to define several stages, for example @keyframes slide { from { left: 0; } to { left: 200px; } } moves an element, and animation: slide 2s infinite applies it for two seconds repeating forever. The animation properties include the keyframe name, duration, timing function, delay, iteration count (a number or 'infinite') and direction. Transitions suit simple state changes like hovers; animations suit repeated or multi-stage motion like a bouncing logo or a sliding banner.

Chapter 9: Data Science and Data Gathering

Multiple Choice Questions

1. What is data?

(a) Processed information

(b) Raw facts gathered about things ✓

(c) A collection of numbers only

(d) A list of observed events

Correct answer: Raw facts gathered about things

Multiple Choice Questions

2. Which of the following is an example of qualitative data?

(a) Temperature readings in degrees Celsius

(b) Number of students in a class

(c) Favourite ice cream flavours ✓

(d) Test scores out of 100

Correct answer: Favourite ice cream flavours

Multiple Choice Questions

3. What type of data involves distinct, separate values that are countable?

(a) Nominal Data

(b) Ordinal Data

(c) Discrete Data ✓

(d) Continuous Data

Correct answer: Discrete Data

Multiple Choice Questions

4. What is an example of continuous data?

(a) Number of cars in a parking lot

(b) Height of students in centimetres ✓

(c) Types of fruits

(d) Shirt sizes (small, medium, large)

Correct answer: Height of students in centimetres

Multiple Choice Questions

5. What type of data is used to categorize items without implying any order?

(a) Ordinal Data

(b) Discrete Data

(c) Nominal Data ✓

(d) Continuous Data

Correct answer: Nominal Data

Multiple Choice Questions

6. How can you organise data to make it easier to analyse?

(a) By writing it in long paragraphs

(b) By creating tables, charts, and graphs ✓

(c) By storing it in random files

(d) By keeping it in a messy notebook

Correct answer: By creating tables, charts, and graphs

Multiple Choice Questions

7. Which tool can be used to create surveys online?

(a) Microsoft Word

(b) Google Forms ✓

(c) Excel Spreadsheets

(d) Adobe Photoshop

Correct answer: Google Forms

Multiple Choice Questions

8. What is the main purpose of data collection?

(a) To create random numbers

(b) To gather information to answer questions or make decisions ✓

(c) To delete old data

(d) To format text documents

Correct answer: To gather information to answer questions or make decisions

Multiple Choice Questions

9. What is the primary purpose of data visualization?

(a) To generate random numbers

(b) To convert text into data

(c) To make data easier to understand by turning it into pictures ✓

(d) To hide complex data

Correct answer: To make data easier to understand by turning it into pictures

Multiple Choice Questions

10. Which tool is specifically designed for creating detailed and interactive visualizations?

(a) Microsoft Excel

(b) Google Sheets

(c) Tableau ✓

(d) PowerPoint

Correct answer: Tableau

Multiple Choice Questions

11. What is the first step in the data science process?

(a) Data Cleaning

(b) Data Collection

(c) Data Analysis

(d) Understanding the problem ✓

Correct answer: Understanding the problem

Multiple Choice Questions

12. What does the 'Volume' characteristic of Big Data refer to?

(a) The speed at which data is generated

(b) The different forms data can take

(c) The sheer amount of data being collected ✓

(d) The way data is processed

Correct answer: The sheer amount of data being collected

Multiple Choice Questions

13. What is an outlier in a dataset?

(a) The most frequent value

(b) The average of all values

(c) An unusual or extreme value that doesn't fit the pattern ✓

(d) The middle value when all values are arranged in order

Correct answer: An unusual or extreme value that doesn't fit the pattern

Multiple Choice Questions

14. What does data encryption do?

(a) It converts data into a code to prevent unauthorized access. ✓

(b) It makes data available to everyone online.

(c) It automatically deletes old data.

(d) It speeds up internet connection.

Correct answer: It converts data into a code to prevent unauthorized access.

Short Questions

1. What is the difference between qualitative and quantitative data?

Qualitative data describes qualities or categories that cannot be measured with numbers, such as favourite ice cream flavours, colours or opinions. Quantitative data is numerical and can be measured or counted, such as temperature readings, test scores or the number of students in a class. In short: qualitative data answers 'what kind?', quantitative data answers 'how much?' or 'how many?'.

Short Questions

2. Give an example of continuous data and explain why it is considered continuous.

Height of students in centimetres is continuous data. It is continuous because it can take any value within a range, including fractions — a student can be 152 cm, 152.5 cm or 152.53 cm tall, with no gaps between possible values. This contrasts with discrete data, such as the number of students, which can only be whole numbers and cannot be 25.5 students.

Short Questions

3. Which method would you use to collect opinions from a large group of people about a new school policy?

A survey, preferably an online survey using a tool such as Google Forms. A survey can reach a large group quickly and cheaply, asks everyone the same questions so answers can be compared, and collects responses in a structured form that is easy to organize into tables and charts for analysis. Interviews would be too slow for a large group, and observation would not capture opinions at all.

Short Questions

4. What type of data is the number of students in your class?

It is quantitative, discrete data. It is quantitative because it is numerical (a count), and discrete because it involves distinct, separate values that are countable: the class can have 30 or 31 students, but never 30.5 students. It is not continuous, since fractional values are impossible, and not nominal or ordinal, since it is not a category or a ranking.

Short Questions

5. Why is it important to organise data into tables or charts before analyzing it?

Organizing data into tables, charts and graphs makes it easier to analyse because raw data in long paragraphs or random files hides patterns. A table arranges values in rows and columns so they can be compared and sorted; a chart turns numbers into pictures so trends, highest and lowest values, and unusual values stand out at a glance. Without organization, analysis is slow and errors are likely; with it, conclusions can be drawn accurately and communicated clearly.

Short Questions

6. What is one advantage of using online tools like Google Forms for collecting survey data?

One major advantage is automatic collection and organization: responses are gathered online from many people at once, without printing or manual entry, and the tool automatically compiles them into a table and summary charts. This saves time, reduces transcription errors, reaches respondents anywhere, and lets the collector analyze results immediately as responses arrive.

Short Questions

7. Why might you need to integrate data from different sources when working on

a project?

A single source rarely tells the whole story, so integrating data from different sources gives a fuller, more accurate picture. For example, a project on school performance might combine exam marks, attendance records and survey opinions: marks alone show results, attendance explains some of them, and opinions reveal causes. Integration also lets you cross-check findings — if two sources agree, the conclusion is more trustworthy — and fills gaps that any one source leaves.

Short Questions

8. Describe a scenario where discrete data might be more useful than continuous data.

Counting attendance is a scenario where discrete data is more useful: a school needs to know exactly how many students are present each day (for example 28, 30 or 31), and fractional values would be meaningless. Discrete counts are exact and easy to compare day by day, to total over a month, and to use for decisions such as how many desks or books are needed. Continuous measurements, such as the average height of students, could not answer the question 'how many students are here today?' at all.

Short Questions

9. Explain why data visualization is important. How does it help in understanding complex information?

Data visualization is important because it turns complex data into pictures — charts, graphs and dashboards — that the human eye can grasp far faster than rows of numbers. A line graph instantly shows whether marks are rising or falling over time; a bar chart instantly shows which subject scored highest; a pie chart shows shares of a whole. Visualization reveals patterns, trends and outliers

(unusual values) that are invisible in raw tables, and it helps decision-makers understand and communicate findings quickly and accurately.

Short Questions

10. Describe what a line graph is used for and provide an example of data that could be displayed using a line graph.

A line graph is used to show how a value changes over time (or another continuous sequence), by plotting points and joining them with lines so trends are visible. For example, a student's marks in monthly tests across a school year could be displayed as a line graph: each month is a point, and the rising or falling line shows whether performance is improving, declining or staying steady. It is the right choice when the order of values matters and change over time is the story.

Short Questions

11. Explain the use of scatter plots in visualizing continuous data. Provide an example of a situation where a scatter plot would be useful.

A scatter plot displays pairs of continuous values as dots on two axes, one value on each axis, so the relationship between the two variables can be seen. It is useful when you want to know whether two things are related: for example, plotting students' study hours (x-axis) against their test scores (y-axis) shows whether more study is associated with higher scores — dots rising left to right suggest a positive relationship, a cloud with no pattern suggests none, and a dot far from the rest reveals an outlier worth investigating. Unlike a line graph, it does not assume an order over time; it reveals correlation between two continuous variables.

Long Questions

1. Explain the differences between qualitative and quantitative data. Provide examples of each type.

Qualitative data describes qualities, categories or opinions that are not measured numerically: favourite ice cream flavours, types of fruits, shirt sizes in words, or answers to 'why do you like this subject?'. It is collected through interviews, open survey questions and observations, and analyzed by grouping responses into themes. Quantitative data is numerical and measured or counted: temperature readings, test scores, heights, or the number of students in a class. It divides into discrete data (countable whole values, such as number of cars in a parking lot) and continuous data (any value in a range, such as height in centimetres). Quantitative data is analyzed with statistics — averages, totals, charts — while qualitative data explains reasons and meanings behind the numbers. A complete study often uses both: numbers show how many, qualities explain why.

Long Questions

2. Describe the process of conducting a survey to gather data about students' favourite extracurricular activities.

Step 1: define the purpose — to learn which extracurricular activities students prefer, so the school can offer the right clubs. Step 2: design the survey — short, clear questions, such as a list of activities to tick (sports, debating, art, science club) plus one open question for suggestions, created in a tool such as Google Forms. Step 3: choose the respondents — for example, all students of Classes 9 and 10, or a representative sample. Step 4: distribute the survey and collect responses, giving a clear deadline and reminding students to answer honestly. Step 5: organize the data into tables and charts, counting how many students chose each activity. Step 6: analyze and report — identify the most and least popular activities, note any patterns (for example by class or gender), and

recommend which clubs to start or expand. Throughout, responses should be kept anonymous so students answer freely.

Long Questions

3. Compare and contrast continuous and discrete data. Use examples to show how each type of data might be used in a school setting, such as in measuring student performance or tracking attendance.

Discrete data involves distinct, separate, countable values, always whole numbers: the number of students present each day, the number of books borrowed, or the number of questions answered correctly. Continuous data can take any value within a range, including fractions: a student's height in centimetres, the time taken to run 100 metres, or the exact percentage score on a test (for example 87.5%). In a school, attendance tracking uses discrete data — you count heads, and 30.5 students is impossible — while measuring performance over time often uses continuous data, such as average test scores with decimals, which can reveal small improvements a whole-number count would hide. Both are quantitative, but discrete data suits counting, continuous data suits measuring, and choosing the wrong type leads to nonsense (a fractional attendance) or lost detail (rounding a precise score).

Long Questions

4. Analyse the benefits and challenges of using digital tools like Google Forms for data collection.

Benefits: digital tools collect data fast and cheaply from many people at once, anywhere; responses are automatically organized into tables and summary charts, reducing manual entry errors; questions can be standardized so answers are comparable; and results can be monitored live as responses arrive.

Challenges: not everyone has reliable Internet access or a device, which can

exclude some respondents and bias results; respondents may rush or misunderstand questions without an interviewer to clarify; duplicate or fake responses are possible; and personal data collected online needs protection, since a breach exposes respondents' information. Overall, digital tools are excellent for large, structured surveys, but they work best when combined with clear question design, access arrangements for those offline, and care for privacy and data security.

Long Questions

5. Imagine you are tasked with organizing a school event and need to collect data on students' preferences for activities and refreshments.

To organize the event well, collect preference data before deciding anything. Step 1: design a short survey (Google Forms) asking students to choose preferred activities (games, quiz, drama, sports) and refreshments (juice, sandwiches, fruit), with an option for allergies or dietary needs. Step 2: distribute it to all students likely to attend, with a deadline, and remind them to respond. Step 3: organize responses into tables and charts: count votes for each activity and refreshment, and separate results by class if needed. Step 4: analyze: the most-voted activities become the main programme, refreshments are ordered in quantities matching the votes (with extra allowance for popular items), and dietary needs are noted for safe catering. Step 5: act and report: publish the chosen programme, and after the event, compare attendance and satisfaction with the survey predictions to improve the next event. This data-driven approach avoids guessing and ensures the event matches what students actually want.

Long Questions

6. Explain the role of tables and charts in data analysis. Provide an example of how you could use a table or chart to present data about students' grades in

different subjects.

Tables and charts are the tools that turn raw data into analyzable, communicable information. A table organizes data into rows and columns so exact values can be looked up, sorted and compared: for example, a table of a class's grades with subjects as columns (English, Mathematics, Science, Urdu) and students as rows lets a teacher find any student's grade in any subject instantly and compute averages per subject. A chart turns the same data into a picture: a bar chart of average grades per subject instantly shows which subject is strongest and which needs attention, and a line graph of one student's grades over the year shows progress or decline. Tables give precision, charts give patterns; together they let analysts draw accurate conclusions and present them so others understand immediately.

Long Questions

7. Describe a situation where non-numeric data is essential. How would you collect, store, and analyse this type of data? Discuss the tools and techniques you would use to ensure the data is accurately interpreted.

Non-numeric (qualitative) data is essential when numbers cannot capture the question — for example, understanding why students dislike a subject, or what parents think of a new school policy. Collection: use open-ended survey questions, interviews and focus-group discussions, recording answers in writing or audio (with permission). Storage: transcribe responses into a document or spreadsheet, one row per respondent, keeping personal details separate and secure. Analysis: read all responses, code them by grouping similar answers into themes (for example 'boring lessons', 'difficult homework', 'lack of practical work'), count how often each theme appears, and quote representative answers to illustrate each theme. To ensure accurate interpretation: use clear, neutral questions that do not lead respondents; have two people code the themes

independently and compare; check findings against quantitative data (if most students also score low in the subject, the themes gain support); and report both the themes and their frequencies, so conclusions rest on evidence rather than a few striking quotes.

Long Questions

8. Explain the concept of data visualization. How does it help in understanding complex data? Provide examples of different types of visualizations and their applications in real-life scenarios.

Data visualization is the practice of turning data into pictures — charts, graphs, maps and dashboards — so complex information can be understood quickly. It helps because the human visual system detects patterns, trends and outliers in a picture far faster than in a table of numbers: a rising line shows growth, a tall bar shows a leader, a distant dot shows an outlier. Examples: a bar chart comparing monthly sales of a shop shows the best month instantly; a line graph of a patient's temperature over days shows fever trends to a doctor; a pie chart of a school budget shows the share spent on books, salaries and maintenance; a scatter plot of study hours versus scores reveals whether study correlates with success; and a dashboard combining several charts lets a business manager monitor sales, stock and complaints at a glance. Tools range from spreadsheets (Excel, Google Sheets) for simple charts to Tableau for detailed, interactive visualizations. Good visualization does not hide complexity — it makes it visible.

Long Questions

9. Discuss the importance and benefits of data visualization. Why is it essential for businesses and decision-makers to use charts, graphs, and dashboards?

Data visualization is essential for decision-makers because decisions must be fast and accurate, and raw data is too large and complex to read directly. Its

benefits: speed — a dashboard shows sales, stock and complaints at a glance, where tables would take hours to read; pattern detection — trends, seasonal cycles and outliers become visible, such as a sudden sales drop or a product's steady rise; communication — a chart convinces a board or a customer far more effectively than a spreadsheet, since everyone can see the same story; and error detection — an outlier dot or an impossible bar stands out visually and prompts investigation. For businesses, this means earlier warnings, better-targeted actions (restocking a rising product, fixing a falling service) and decisions based on evidence rather than guesswork. Without visualization, data remains collected but unused; with it, data becomes decisions.

Long Questions

10. Differentiate between nominal, ordinal, discrete, and continuous data. For each type, describe a suitable visualization technique and provide a specific example of how this technique can be used to represent that type of data effectively.

Nominal data categorizes items without any order — for example, types of fruits or favourite ice cream flavours. Suitable visualization: a bar chart or pie chart, for example a bar chart showing how many students prefer each ice cream flavour, where the order of bars carries no meaning. Ordinal data categorizes with a meaningful order or ranking — for example, shirt sizes (small, medium, large) or satisfaction ratings (poor, fair, good, excellent). Suitable visualization: an ordered bar chart, for example bars arranged from 'poor' to 'excellent' showing survey responses, where the order is part of the message. Discrete data is countable whole numbers — for example, the number of students in each class. Suitable visualization: a bar chart or column chart, for example columns showing class sizes, where each column is an exact count. Continuous data can take any value in a range — for example, heights of students in centimetres.

Suitable visualization: a histogram (bars over measurement intervals) or a line graph over time, for example a histogram of student heights showing how many fall in each height band, which reveals the distribution's shape. Choosing the matching visualization prevents misleading pictures, such as implying an order that does not exist or hiding gaps between values.

Chapter 10: Emerging Technologies in Computer Science

Multiple Choice Questions

1. Which of the following is not a subfield of AI? (a) Machine Learning (b) Natural Language Processing (c) Computer Vision (d) Robotics — NOTE on the question as printed

As printed, this question has no correct option, because all four options are recognized subfields of Artificial Intelligence: machine learning (systems that learn from data), natural language processing (computers understanding human language), computer vision (computers seeing and understanding the visual world), and robotics (building and programming robots). None of them is outside AI, so a student should note the error in the book rather than guess.

Multiple Choice Questions

2. Which of these AI algorithms is considered an 'explainable' model?

(a) Neural Networks

(b) Decision Trees ✓

(c) Random Forests

(d) Convolutional Neural Networks

Correct answer: Decision Trees

Multiple Choice Questions

3. Which of these is a security concern in IoT deployments?

(a) Device vulnerability

(b) Data privacy

(c) Lack of standardization

(d) All of the above ✓

Correct answer: All of the above

Multiple Choice Questions

4. Which of the following is an application of AI in healthcare?

(a) Personalized drug development

(b) Automated diagnosis

(c) Remote patient monitoring

(d) All of the above ✓

Correct answer: All of the above

Multiple Choice Questions

5. What is the primary purpose of using AI techniques in machine learning models?

(a) To improve accuracy

(b) To enhance interpretability

(c) To reduce computational complexity

(d) All of the above ✓

Correct answer: All of the above

Multiple Choice Questions

6. What is the key difference between explainable (whitebox) and unexplainable (blackbox) AI models?

(a) The complexity of the model

(b) The ability to understand the decision-making process ✓

(c) The performance of the model

(d) The training data used

Correct answer: The ability to understand the decision-making process

Multiple Choice Questions

7. Which of the following is an application of IoT in the transportation domain?

(a) Smart traffic management

(b) Vehicle-to-Vehicle (V2V) communication

(c) Predictive maintenance of vehicles

(d) All of the above

Correct answer: All of the above

Multiple Choice Questions

8. Which of these is a potential impact of AI and IoT on the job market?

(a) Job displacement due to automation

(b) Increased demand for specialized skills

(c) Transformation of job roles and responsibilities

(d) All of the above

Correct answer: All of the above

Multiple Choice Questions

9. What is the key concern associated with algorithmic bias in AI-powered decision-making processes?

(a) Lack of transparency

(b) Perpetuation of existing societal biases

(c) Reduced accuracy of the model

(d) All of the above

Correct answer: All of the above

Multiple Choice Questions

10. Which of the following is an ethical principle that should be considered in the development and deployment of AI and IoT technologies?

(a) Transparency and accountability

(b) Respect for privacy and data rights

(c) Fairness and non-discrimination

(d) All of the above

Correct answer: All of the above

Short Questions

1. Define Artificial Intelligence (AI).

Artificial Intelligence is a rapidly growing field of computer science that builds systems able to perform tasks that normally require human intelligence — such as understanding language, recognizing images, making decisions and learning from experience. Examples range from voice assistants and recommendation systems to automated diagnosis in healthcare and self-driving vehicles.

Short Questions

2. What is the historical context and evolution of AI?

AI began as an academic idea in the mid-20th century, when researchers asked whether machines could think and built early programs for reasoning and problem-solving. Progress was slow at first due to limited computing power and data, but faster hardware, large datasets and new techniques such as machine learning and deep learning produced rapid advances from the 2000s onward. Today AI is embedded in everyday life — in phones, search engines, healthcare, transport and industry — and continues to evolve toward more capable and more autonomous systems.

Short Questions

3. Provide two examples of AI applications in healthcare.

Two examples are automated diagnosis, where AI systems analyze medical images or patient data to detect diseases such as cancers or eye conditions, often faster and with high accuracy; and remote patient monitoring, where

wearable devices and AI track a patient's vital signs continuously and alert doctors to worrying changes. A third example from the book is personalized drug development, where AI helps design treatments tailored to individual patients.

Short Questions

4. Explain the role of AI techniques in advancing machine learning models.

AI techniques advance machine learning models by improving their accuracy, enhancing their interpretability and reducing their computational complexity. Better algorithms learn more accurate patterns from data; techniques such as decision trees make models explainable so users can understand their decisions; and optimization methods let models train faster and run on smaller devices. Together these advances turn raw machine learning into practical, trustworthy systems.

Short Questions

5. Define the Internet of Things (IoT).

The Internet of Things (IoT) is the network of physical objects — 'things' — embedded with sensors, software and other technologies that connect and exchange data with other devices and systems over the Internet. Examples include smart thermostats, fitness trackers, connected vehicles and industrial sensors, all of which sense their surroundings, communicate data, and act on it without constant human control.

Short Questions

6. Describe the significance of IoT in connecting devices and systems.

IoT is significant because it connects previously isolated devices and systems into one communicating network, so data flows automatically between them. A smart home's lights, thermostat and security cameras can coordinate; a

factory's machines can report their own condition; and a city's traffic lights can respond to real traffic. This connectivity enables automation, remote monitoring and control, and data-driven decisions that would be impossible if each device worked alone.

Short Questions

7. What are the potential risks associated with AI and IoT?

The main risks are security and privacy risks: IoT devices can be vulnerable to hacking (device vulnerability), personal data collected by devices can leak or be misused (data privacy), and the lack of common standards makes consistent protection difficult. AI adds the risks of algorithmic bias (unfair decisions), lack of transparency in blackbox models, and job displacement as automation replaces some roles. Mitigating these risks needs secure design, privacy safeguards, standards and regulation.

Short Questions

8. Discuss the societal impact of AI and IoT on daily life.

AI and IoT have transformed daily life: voice assistants, recommendations, navigation and smart home devices make routine tasks easier; healthcare, education and transport benefit from faster information and automation; and businesses operate more efficiently. At the same time they raise concerns: automation displaces some jobs while creating demand for new specialized skills, constant data collection threatens privacy, and dependence on connected systems creates new vulnerabilities. Society therefore gains convenience and productivity, but must manage fairness, privacy and employment transitions.

Short Questions

9. Explain the concept of algorithmic bias.

Algorithmic bias occurs when an AI system systematically produces unfair outcomes for particular groups of people. It usually comes from biased training data — for example, if historical hiring data favours one group, a model trained on it will reproduce that favouritism — or from design choices that overlook some groups. Its key concern is the perpetuation of existing societal biases, now hidden inside an apparently objective machine decision, which can affect hiring, lending, policing and healthcare. It is worsened by lack of transparency in blackbox models, which makes the bias hard to detect.

Short Questions

10. Outline the importance of ethical considerations in AI and IoT.

Ethical considerations are important because AI and IoT systems make decisions affecting people's health, livelihoods, privacy and freedom, so they must be developed and deployed responsibly. Key principles are transparency and accountability (decisions must be explainable and someone must answer for them), respect for privacy and data rights (personal data must be collected fairly and protected), and fairness and non-discrimination (systems must not perpetuate bias). Without these principles, powerful technologies can harm the very people they are meant to serve, and public trust in them collapses.

Long Questions

1. Discuss the various applications of AI in the field of education. Provide specific examples and explain how AI can enhance the educational experience.

AI enhances education in several ways. Personalized learning: adaptive platforms analyze each student's strengths and weaknesses and adjust lessons, exercises and pace to the individual, so a struggling student gets extra practice while an advanced student moves ahead. Intelligent tutoring: AI tutors answer questions, explain concepts and give instant feedback outside class hours.

Automated assessment: AI marks objective tests quickly and consistently, freeing teachers for creative work, and can flag students at risk from their performance patterns. Accessibility: speech recognition, translation and text-to-speech help students with disabilities or language barriers. Administration: AI handles scheduling, attendance and routine queries, letting teachers focus on teaching. Together these make learning more personal, more accessible and more responsive, though teachers remain essential for guidance, motivation and judgement that machines cannot provide.

Long Questions

2. Differentiate between explainable (whitebox) and unexplainable (blackbox) AI models.

Whitebox (explainable) models are those whose decision-making process is transparent and understandable: a human can follow how inputs lead to the output. Decision trees are the classic example — each decision is a visible rule (for example, 'if marks are 90 or above, grade A+'). Their advantage is trust and auditability: errors and biases can be found and explained, which matters in healthcare, finance and law. Blackbox (unexplainable) models, such as neural networks and convolutional neural networks, are those whose decision-making process is not easily interpretable: they may be highly accurate, but even their creators cannot fully explain a particular decision, since it emerges from thousands of weighted connections. Their advantage is performance on complex tasks such as image recognition; their disadvantage is opacity, which makes bias, errors and accountability hard to handle. The choice is a trade-off between accuracy and explainability, and sensitive decisions increasingly demand whitebox or explainable approaches.

Long Questions

3. Describe the components of an IoT system. Explain how these components work together to enable IoT applications.

An IoT system has four main components. Sensors and devices collect data from the environment — for example, a temperature sensor, a camera or a fitness tracker's heart-rate monitor. Connectivity carries that data over a network, using Wi-Fi, Bluetooth, mobile networks or wired links, to where it can be processed. Data processing and platforms (often cloud services) store and analyze the data, applying rules or AI models to decide what it means — for example, that a room is too warm. Actuators and applications act on the decision: the system switches on the air conditioner, sends an alert to the user's phone, or displays a dashboard. Working together: the sensor senses, the network transports, the platform decides, and the actuator acts, creating a loop that runs with little human involvement — the essence of a smart home, a connected vehicle or a smart factory.

Long Questions

4. Explore the applications of IoT in the transportation domain.

IoT transforms transportation in several ways. Smart traffic management: road sensors and cameras measure traffic flow in real time, and signals adjust automatically to reduce congestion. Vehicle-to-Vehicle (V2V) communication: vehicles exchange speed, position and hazard warnings directly, helping prevent collisions. Predictive maintenance: sensors on vehicles and infrastructure monitor engine condition, tyre pressure and track wear, predicting failures before they happen so maintenance is scheduled rather than emergency. Fleet and logistics tracking: connected vehicles report location and cargo condition, letting companies route deliveries efficiently. Public transport: buses and trains report real-time positions so passengers see accurate arrival times. Together these make transport safer, more efficient and less polluting.

Long Questions

5. Analyze the potential privacy concerns associated with IoT deployments.

IoT deployments create serious privacy concerns because they collect continuous, detailed data about people's lives. Devices in homes, on bodies and in vehicles record location, habits, conversations, health signs and routines, often without users fully realizing how much is gathered. Risks include: data breaches, where hacked devices leak personal information; secondary use, where data collected for one purpose (for example, a fitness tracker) is sold or used for advertising or insurance decisions without clear consent; surveillance, where many devices together can reconstruct a person's movements and behaviour; and weak security, since many small devices lack strong protection or updates. Mitigations include collecting only necessary data, encrypting data in transit and storage, clear consent and privacy settings, regular security updates, and laws that hold providers accountable for protecting personal data.

Long Questions

6. Evaluate the impact of AI and IoT on the job market and work environments.

AI and IoT affect the job market in three ways. Job displacement: automation replaces routine, repetitive tasks — for example, data entry, basic manufacturing and routine customer queries — putting some roles at risk. Increased demand for specialized skills: new roles appear in AI development, data analysis, cybersecurity and IoT maintenance, rewarding workers with technical skills. Transformation of job roles: most jobs change rather than vanish, as workers use AI tools to work faster and focus on creative, interpersonal and judgement-based tasks that machines cannot do. Work environments also change: remote monitoring, smart offices and AI-assisted decisions become normal, while employers must retrain staff and manage the transition fairly. The net effect depends on policy and training: with investment in education, AI and IoT can

raise productivity and create better jobs; without it, they can widen inequality.

Long Questions

7. Explain the role of policy and regulatory frameworks in addressing the challenges of AI and IoT. Provide examples of existing frameworks and discuss their effectiveness.

Policy and regulatory frameworks set the rules that keep AI and IoT safe, fair and accountable, since technology alone does not guarantee responsible use. They address challenges in several areas: data protection laws (such as Pakistan's Personal Data Protection Bill, discussed in Chapter 11, and Europe's GDPR) require consent, limit collection and give users rights over their data; safety and product standards require devices to meet security baselines before sale; anti-discrimination rules require that automated decisions do not unfairly harm protected groups; and sector rules govern high-risk uses such as medical AI and autonomous vehicles. Their effectiveness is mixed: strong frameworks raise standards and give users remedies, but they struggle to keep pace with fast-moving technology, to enforce rules across borders, and to cover small, cheap IoT devices made by many manufacturers. Effective regulation therefore combines law with standards, industry responsibility and public awareness, and must be updated as technology evolves.

Long Questions

8. Describe the concept of algorithmic bias and its implications in AI-powered decision-making processes. Suggest strategies to mitigate the risks of algorithmic bias.

Algorithmic bias is systematic unfairness in an AI system's decisions, usually inherited from biased training data or from design choices that overlook some groups, with the key concern that it perpetuates existing societal biases behind

an apparently objective machine. Its implications are serious: a biased hiring tool can reject qualified candidates from one group; a biased lending model can deny credit unfairly; biased policing or healthcare tools can harm communities already disadvantaged. Mitigation strategies: use diverse, representative training data and audit it for gaps; test models for fairness across groups before deployment and monitor them afterwards; prefer explainable (whitebox) models for high-stakes decisions so bias can be seen; include diverse teams in design and review; provide human oversight and an appeal route for affected people; and follow fairness principles and regulations that require accountability. No single fix removes bias, but this combination greatly reduces it.

Long Questions

9. Develop a set of ethical principles and guidelines for the responsible development and deployment of AI and IoT technologies.

A responsible set of principles is: (1) Transparency and accountability — systems must explain their decisions in understandable terms, and a named person or organization must answer for their outcomes. (2) Privacy and data rights — collect only necessary data, obtain informed consent, encrypt data, and respect users' rights to access and delete it. (3) Fairness and non-discrimination — test for bias across groups, use representative data, and correct unfair outcomes. (4) Safety and security — design devices and models to be secure by default, update them against new threats, and test them thoroughly before release. (5) Human oversight — keep humans in control of high-stakes decisions, with the power to override the system. (6) Beneficence — deploy AI and IoT to serve human well-being, such as health, education and sustainability, not merely profit. (7) Legal compliance — follow applicable laws, including data protection and intellectual property law. Guidelines for practice: conduct ethics and impact assessments before deployment, document data sources and model limits,

monitor systems after release, report incidents openly, and train developers and users in these principles.

Chapter 11: Ethical, Social, and Legal Concerns in Computer Usage

Multiple Choice Questions

1. Why is it important to use computers safely and responsibly?

(a) To ensure we can use them more frequently

(b) To protect our personal information and make wise choices about hardware and software ✓

(c) To make the computer run faster

(d) To avoid paying for software

Correct answer: To protect our personal information and make wise choices about hardware and software

Multiple Choice Questions

2. What does 'responsible computer usage' include?

(a) Choosing the most expensive hardware

(b) Sharing your passwords with friends

(c) Being cautious about what you share online and protecting yourself and others ✓

(d) Ignoring software updates

Correct answer: Being cautious about what you share online and protecting yourself and others

Multiple Choice Questions

3. What should you check to ensure hardware and software compatibility?

(a) The color of the hardware

(b) The system requirements on software packages and match them with your computer's specifications ✓

(c) The price of the hardware

(d) The brand of the hardware

Correct answer: The system requirements on software packages and match them with your computer's specifications

Multiple Choice Questions

4. Why is it important to use strong, unique passwords?

(a) To make your accounts easy to hack

(b) To help others guess your password

(c) To make it harder for someone to guess your password and access your accounts

(d) To avoid installing software updates

Correct answer: To make it harder for someone to guess your password and access your accounts

Multiple Choice Questions

5. What is one reason to avoid clicking on unknown links or downloading files from untrusted sources?

(a) They might contain helpful software

(b) They might contain malware that can harm your device or steal your information

(c) They are usually cheaper

(d) They help your device run faster

Correct answer: They might contain malware that can harm your device or steal your information

Multiple Choice Questions

6. What does Two-Factor Authentication (2FA) do?

(a) It makes your password easier to guess

(b) It adds an extra layer of security by requiring a second form of verification

(c) It removes the need for a password

(d) It reduces the need for software updates

Correct answer: It adds an extra layer of security by requiring a second form of verification

Multiple Choice Questions

7. Why should you be cautious when using public Wi-Fi for sensitive transactions?

(a) Public Wi-Fi is usually faster

(b) Public Wi-Fi networks are often less secure

(c) Public Wi-Fi is free

(d) Public Wi-Fi always provides encryption

Correct answer: Public Wi-Fi networks are often less secure

Multiple Choice Questions

8. What is an important aspect of responsible use of social media?

(a) Sharing personal information like your home address

(b) Posting photos without considering privacy settings

(c) Being respectful and avoiding sharing sensitive information publicly

(d) Ignoring privacy settings

Correct answer: Being respectful and avoiding sharing sensitive information publicly

Multiple Choice Questions

9. What should you do if you receive an email from an unknown sender asking for personal information?

(a) Provide the information requested

(b) Forward the email to your friends

(c) Ignore or delete the email ✓

(d) Open the email and click on any links

Correct answer: Ignore or delete the email

Multiple Choice Questions

10. Why is it important to regularly review your account activity?

(a) To update your contact information

(b) To look for unusual activity and ensure your accounts are secure ✓

(c) To check the number of friends you have

(d) To download new applications

Correct answer: To look for unusual activity and ensure your accounts are secure

Multiple Choice Questions

11. What is the purpose of privacy settings on digital platforms?

(a) To make your posts public

(b) To control who can see your information and interact with you online ✓

(c) To increase the number of followers

(d) To automatically share your information

Correct answer: To control who can see your information and interact with you online

Multiple Choice Questions

12. What should you do to ensure data security?

(a) Use the same password for all accounts

(b) Share your passwords with friends

(c) Use strong, unique passwords and enable two-factor authentication ✓

(d) Avoid using any security measures

Correct answer: Use strong, unique passwords and enable two-factor authentication

Multiple Choice Questions

13. What is a key aspect of data ethics?

(a) Using data in any way you like

(b) Transparency, respect for privacy, and accountability in data usage ✓

(c) Collecting as much data as possible

(d) Ignoring data security

Correct answer: Transparency, respect for privacy, and accountability in data usage

Multiple Choice Questions

14. What is software piracy?

(a) Sharing software legally with friends

(b) The illegal copying, distribution, or use of software ✓

(c) Buying software from an official source

(d) Updating software regularly

Correct answer: The illegal copying, distribution, or use of software

Multiple Choice Questions

15. How can you verify the credibility of information found online?

(a) By checking the number of ads on the website

(b) By using multiple reputable sources and checking the author's credentials ✓

(c) By looking at the website's design

(d) By the website's popularity

Correct answer: By using multiple reputable sources and checking the author's credentials

Short Questions

1. What is the importance of using computers safely and responsibly?

Using computers safely and responsibly is important to protect your personal information and to make wise choices about hardware and software. Safe use prevents malware infections, identity theft and data loss; responsible use — respecting others, obeying the law and protecting privacy — prevents harm to other people and keeps you on the right side of rules and laws. In short, safe and responsible use lets you gain the benefits of computing without exposing yourself or others to its risks.

Short Questions

2. How does choosing the right hardware and software affect your computer usage?

Choosing the right hardware and software determines whether your computer works well at all: compatible, adequate hardware runs software smoothly, while underpowered or incompatible hardware causes crashes, slowness and wasted money; and suitable software — matched to your needs and your system's requirements — performs its tasks correctly and securely. Good choices improve safety (supported software receives security updates), efficiency (tasks complete quickly) and compatibility (devices and programs work together); poor choices cause frustration, faults and security gaps.

Short Questions

3. Why is it important to use antivirus software on your computer?

Antivirus software is important because it detects, quarantines and removes malware — viruses, spyware and other harmful programs — that can steal personal information, corrupt files, slow the system or turn the computer into a tool for attacking others. Since new malware appears constantly, antivirus

software must be kept updated so it recognizes the latest threats; without it, a single careless click or download can compromise the whole system.

Short Questions

4. What are some examples of good practices when selecting hardware and software?

Good practices include: checking the system requirements on the software package and matching them with your computer's specifications before buying; choosing reputable brands and official sources rather than unknown downloads; preferring hardware and software that will receive updates and support; considering your real needs (for example, a student needs a word processor, not expensive professional design software); and checking compatibility between new hardware, existing devices and the operating system. These practices save money, prevent faults and improve security.

Short Questions

5. How can choosing compatible hardware and software enhance your computer experience?

Compatible hardware and software enhance your experience because everything works together smoothly: programs install and run without errors, devices are recognized immediately, performance is stable, and features work as advertised. Incompatible choices cause the opposite — crashes, missing features, driver problems and wasted time troubleshooting. Compatibility therefore means less frustration, better performance and a system you can rely on for study and work.

Short Questions

6. Why should you create strong, unique passwords for your accounts?

Strong, unique passwords make it harder for someone to guess your password

and access your accounts. 'Strong' means long and mixed (upper- and lower-case letters, digits and symbols) so guessing programs cannot crack it quickly; 'unique' means a different password for each account, so if one account is breached, the attacker cannot use the same password to enter your email, bank and social media accounts as well. Weak or reused passwords are the commonest cause of account takeovers.

Short Questions

7. What is the purpose of regular software updates?

Regular software updates fix bugs, improve performance and, most importantly, close security vulnerabilities that attackers could exploit. An un-updated operating system or application stays exposed to known attacks, while an updated one is protected against them. Updates may also add features and improve compatibility with new hardware, so keeping software current is one of the simplest and most effective security practices.

Short Questions

8. How can you protect yourself from harmful links and downloads?

Protect yourself by never clicking unknown links or downloading files from untrusted sources, since they may contain malware that harms your device or steals your information. Check the sender and the address carefully, hover over links to see their real destination, download software only from official websites or app stores, keep antivirus software active and updated, and if a message pressures you to act urgently (a common phishing trick), stop and verify through a separate, trusted channel before acting.

Short Questions

9. What is Two-Factor Authentication (2FA), and why is it useful?

Two-Factor Authentication adds an extra layer of security by requiring a second form of verification in addition to the password — typically a code sent to your phone, generated by an authenticator app, or a fingerprint. It is useful because even if an attacker steals or guesses your password, they still cannot enter the account without the second factor, which they do not possess. Enabling 2FA is one of the strongest single steps for protecting important accounts.

Short Questions

10. Why is it a good idea to avoid using public Wi-Fi for sensitive transactions?

Public Wi-Fi networks are often less secure than private ones: they may lack encryption, and attackers on the same network can intercept traffic or set up fake hotspots that imitate a legitimate network. Avoiding sensitive transactions (banking, shopping, entering passwords) on public Wi-Fi prevents your financial and personal details from being captured. If you must use public Wi-Fi, use a VPN, check that sites use HTTPS, and avoid logging into important accounts.

Short Questions

11. How can you verify if an email or message is a scam?

Check for warning signs: an unknown sender, urgent threats or promises (account closure, a prize), requests for personal information or passwords, spelling and grammar errors, and links or attachments you did not expect. Verify independently: do not click the message's links; instead, contact the supposed organization through its official website or phone number, or check your account directly by typing the address yourself. If it asks for personal information from an unknown sender, ignore or delete it and report it as phishing.

Short Questions

12. Why is it important to be careful about what you share on social media?

What you share on social media can be copied, spread and stored far beyond your intended audience, and it is difficult to delete completely. Sharing personal information (home address, school, phone number, daily routine) or sensitive photos exposes you to identity theft, stalking, bullying and scams, and can harm your reputation with future schools and employers. Responsible use means being respectful, using privacy settings to control who sees your posts, and never sharing sensitive information publicly.

Short Questions

13. What should you do if you receive an email from an unknown sender asking for personal information?

Do not provide the information, do not forward the email to friends, and do not click any links in it. The safe response is to ignore or delete the email, and to report it as phishing where your email service offers that option. Legitimate organizations never ask for passwords or sensitive details by email; if you are unsure whether a message is genuine, contact the organization through an official channel you find yourself, not through the message.

Short Questions

14. What is the purpose of privacy laws regarding your personal information?

Privacy laws exist to protect individuals from unauthorized access to and misuse of their personal information. They require organizations to collect data fairly and only for stated purposes, to keep it secure, to let people know what is held about them, and to face consequences for breaches. In Pakistan, the book cites the Personal Data Protection Bill as the framework giving citizens these protections in the digital age.

Short Questions

15. How do privacy laws protect you from unauthorized access to your data?

Privacy laws protect you by placing legal duties on whoever holds your data: they must obtain your consent, use the data only for the purpose you agreed to, protect it with reasonable security measures, and not share or sell it without permission. They give you rights — to know what data is held, to correct it, and to complain — and they punish organizations that access, leak or misuse data without authorization. This legal backing turns good practice into an enforceable obligation.

Short Questions

16. What is the difference between copyright, trademarks, and patents?

Copyright protects original creative works — books, music, films, software and art — giving the creator the exclusive right to copy and distribute them for a set period. A trademark protects brand identifiers — names, logos and slogans, such as a company's logo — that distinguish one business's goods or services from another's. A patent protects inventions — new, useful processes, machines or designs — giving the inventor exclusive rights to make and sell the invention for a limited time. In short: copyright covers expression, trademarks cover brand identity, and patents cover inventions.

Short Questions

17. Why is it important to respect intellectual property rights?

Respecting intellectual property rights is important because creators and inventors deserve to benefit from their work, and that reward encourages further creativity and innovation. Using someone else's work without permission — copying software, music or text — deprives them of income and is illegal, with consequences including fines and legal action. Respect also protects you: using licensed, genuine software is safer (pirated copies often carry malware) and

keeps your own work credible and lawful.

Short Questions

18. What is software piracy, and why is it harmful?

Software piracy is the illegal copying, distribution or use of software — for example, installing one licensed copy on many computers, downloading cracked programs, or selling counterfeit discs. It is harmful to developers and companies, who lose the income that funds new software; to users, since pirated software receives no updates or support and often contains malware; and to the economy, since it undermines legitimate businesses. Respecting licences and buying from official sources avoids these harms and the legal consequences of infringement.

Short Questions

19. How can you identify reliable sources when researching online?

Identify reliable sources by using multiple reputable sources and checking the author's credentials: prefer established institutions, academic publishers, official organizations and recognized experts over anonymous blogs or sponsored pages. Check whether the information is dated, whether it cites its own sources, whether other reputable sources agree, and whether the site's purpose is to inform rather than to sell or mislead. Cross-checking across independent sources is the strongest test: a claim appearing in only one dubious source should not be trusted.

Short Questions

20. What is one way to protect your privacy during online research?

One effective way is to use privacy settings and tools consciously: use a browser's private mode, adjust search-engine and account privacy settings so

your searches are not linked to your identity, avoid logging into personal accounts while researching sensitive topics, and never enter personal details on sites you have not verified. Combined with avoiding public Wi-Fi for sensitive work and keeping software updated, these habits keep your research activity from exposing your identity and interests.

Short Questions

21. What are some signs that you might be developing an internet addiction?

Signs include spending far more time online than intended and being unable to cut back; neglecting studies, sleep, meals, family and offline activities in favour of being online; feeling restless, anxious or irritable when offline; needing ever more time online to feel satisfied; hiding the extent of your use from others; and continuing heavy use despite harm to grades, health or relationships. Recognizing these signs early matters, because setting time limits, finding offline activities and seeking support can restore a balanced pattern of use.

Long Questions

1. Discuss the importance of responsible computer usage in today's digital world. Explain how selecting the right hardware and software can affect safety, efficiency, and compatibility in computer use.

Responsible computer usage means using computers safely and wisely: protecting personal information, making wise hardware and software choices, respecting others online, and obeying the law. It matters more than ever because computers now hold our money, identity, education and social life, so careless use exposes us to malware, fraud and reputational harm, while responsible use keeps us and others safe. Selecting the right hardware and software affects all three dimensions. Safety: compatible, supported hardware and genuine, updated software receive security patches and work with antivirus

protection, while pirated or unsupported software leaves known vulnerabilities open. Efficiency: hardware matched to the task (enough RAM and storage, a suitable processor) and software matched to the need runs smoothly instead of crashing or crawling, saving time. Compatibility: checking system requirements before purchase ensures programs install, devices are recognized and files exchange correctly, instead of driver errors and wasted money. In short, responsible choices at purchase time prevent most of the problems that troubleshooting must later fix.

Long Questions

2. Describe the steps you should take to ensure the safe operation of digital platforms and devices.

Safe operation follows a routine. (1) Choose compatible hardware and software, checking system requirements against your specifications. (2) Create strong, unique passwords for every account and enable two-factor authentication where available. (3) Keep the operating system, applications and antivirus software updated, and run regular antivirus scans. (4) Be cautious with links and downloads: avoid unknown links and untrusted sources, and verify suspicious messages through official channels. (5) Avoid sensitive transactions on public Wi-Fi, which is often less secure, and use HTTPS sites and VPNs when necessary. (6) Use privacy settings to control who sees your information, and be careful what you share on social media. (7) Review account activity regularly for unusual logins or transactions, and act immediately on anything suspicious. (8) Back up important data to external or cloud storage. (9) Respect intellectual property: use licensed software and avoid piracy. Following these steps systematically keeps devices, data and accounts safe.

Long Questions

3. Explain the concept of data ethics and its importance in handling personal and sensitive information. Discuss the principles of transparency, respect for privacy, and accountability.

Data ethics is the branch of ethics concerned with the responsible collection, use and sharing of data, especially personal and sensitive information. It is important because data about people can harm them if misused — through discrimination, exposure or manipulation — and because the people whose data is used rarely control how it is processed. Three principles govern it. Transparency: people should know what data is collected, why, and how it will be used, in clear language rather than hidden terms. Respect for privacy: only necessary data should be collected, with consent, kept secure, and never used beyond the agreed purpose. Accountability: whoever collects and uses data must answer for it — correcting errors, reporting breaches, and accepting consequences for misuse. Together these principles ensure that data serves people rather than exploiting them, and they underpin privacy laws such as Pakistan's Personal Data Protection Bill.

Long Questions

4. Analyze the impact of privacy laws on protecting personal information in the digital age. How do laws like the Personal Data Protection Bill in Pakistan help safeguard user data?

Privacy laws change data protection from a courtesy into an enforceable duty, and their impact is threefold. First, they limit collection: organizations may collect personal data only for stated, lawful purposes and with consent, reducing the mass harvesting that characterizes unregulated data markets. Second, they require protection: holders must secure data against unauthorized access, report breaches, and respect users' rights to know, correct and delete their data. Third, they create accountability: violations can be investigated and punished, giving

users a remedy they lack when protection depends on a company's goodwill. Pakistan's Personal Data Protection Bill, cited in the book, would bring these safeguards to Pakistani users: it would require consent-based collection, restrict cross-border transfer of personal data, oblige organizations to protect data in their care, and give citizens recourse against misuse. Its practical effect depends on enforcement, but even as a framework it sets the standard that responsible organizations must meet.

Long Questions

5. Discuss the different types of intellectual property rights, including copyright, trademarks, and patents.

Intellectual property rights protect creations of the mind in three main forms. Copyright protects original creative expression — books, articles, music, films, photographs and software code — giving the creator exclusive rights to reproduce, distribute and adapt the work, typically for the creator's life plus a set period. It arises automatically when a work is created and does not protect ideas themselves, only their expression. Trademarks protect brand identity — names, logos, slogans and distinctive designs that identify the source of goods or services, such as a company's logo — preventing others from using confusingly similar marks that would mislead customers; trademark rights can last indefinitely while the mark is in use. Patents protect inventions — new, useful and non-obvious processes, machines, manufactures or compositions — granting the inventor exclusive rights to make, use and sell the invention for a limited period (commonly 20 years) in exchange for publicly disclosing how it works. Together they reward creativity, protect consumers from confusion, and encourage innovation by letting creators benefit from their work.

Long Questions

6. Describe the ethical and legal responsibilities related to intellectual property rights. What are the consequences of violating these rights, such as through software piracy or unauthorized use of copyrighted material?

Ethical responsibilities: respect creators' rights by using only licensed or authorized material, crediting sources, and not copying, distributing or altering others' work without permission. Legal responsibilities: obey copyright, trademark and patent law — buy genuine software, respect licence terms (such as the number of permitted installations), and do not reproduce protected works beyond fair-use limits. Consequences of violation include: civil liability — fines and damages payable to the rights holder; criminal penalties in serious cases, such as large-scale piracy; loss of access, for example accounts or services terminated for infringement; security risks, since pirated software often carries malware and receives no updates; and reputational harm for individuals and organizations caught infringing. For students, the practical rule is simple: if you did not create it and do not have permission or a licence, do not copy, share or install it.

Long Questions

7. Outline effective techniques for conducting safe and credible online research. How can users evaluate the reliability of sources and avoid privacy risks during their research?

Safe, credible research combines source evaluation with privacy care. Evaluate reliability by: using multiple reputable sources and checking the author's credentials; preferring official, academic and established institutional sources; checking dates and whether claims cite evidence; cross-checking a claim across independent sources; and watching for bias, advertising disguised as information, and sensational language. Avoid privacy risks by: using privacy settings and private browsing where appropriate; not entering personal details

on unverified sites; avoiding sensitive research on public Wi-Fi; keeping antivirus software and browsers updated; and not downloading files from untrusted sources, which may carry malware. Record your sources as you go, so every claim in your work can be traced back to a credible origin — this both strengthens your work and protects you from relying on misinformation.

Long Questions

8. Discuss the concept of internet addiction and its potential impact on individuals. How can recognizing the signs of addiction, setting time limits, and finding offline activities help promote balanced internet usage?

Internet addiction is a pattern of excessive, compulsive online use that continues despite harm to studies, sleep, health, relationships and work. Signs include losing track of time online, neglecting offline responsibilities, feeling restless or anxious when offline, needing more time online for the same satisfaction, and hiding usage from others. Its impact can be serious: falling grades, sleep deprivation, poor physical health from inactivity, weakened family and social relationships, and anxiety or low mood. Recognizing the signs is the first step to change, because it turns an invisible habit into a problem that can be managed. Setting time limits — scheduled online hours, app timers, device-free meals and bedtimes — restores control and protects sleep and study. Finding offline activities — sports, reading, hobbies, time with family and friends — replaces the rewards the Internet supplied with healthier ones, reducing cravings and rebuilding real-world connections. Together, awareness, limits and alternatives promote balanced usage: the Internet remains a tool for learning and communication, rather than a compulsion that controls the user.

Chapter 12: Entrepreneurship in Digital Age

Multiple Choice Questions

1. What is entrepreneurship?

(a) The process of starting a new business or organization ✓

(b) The process of buying and selling stocks

(c) The process of working for a large company

(d) The process of creating a marketing campaign

Correct answer: The process of starting a new business or organization

Multiple Choice Questions

2. What is a key characteristic of entrepreneurs?

(a) Avoiding risks

(b) Seeking job security

(c) Innovation ✓

(d) Following established methods

Correct answer: Innovation

Multiple Choice Questions

3. What is an example of a digital technology used by modern entrepreneurs?

(a) Newspaper advertisements

(b) Television commercials

(c) Mobile apps ✓

(d) Door-to-door sales

Correct answer: Mobile apps

Multiple Choice Questions

4. Which of the following is an example of a digital tool for creating and editing documents online?

(a) SurveyMonkey

(b) Google Docs ✓

(c) SEMrush

(d) Zendesk

Correct answer: Google Docs

Multiple Choice Questions

5. Which tool is commonly used for optimizing website content for search engines?

(a) Hootsuite

(b) Yoast SEO ✓

(c) Shopify

(d) PayPal

Correct answer: Yoast SEO

Multiple Choice Questions

6. Which of the following is NOT a technique for identifying market needs?

(a) Surveys and Questionnaires

(b) Market Research

(c) Observation

(d) Brainstorming ✓

Correct answer: Brainstorming

Multiple Choice Questions

7. What is the first step in the design thinking process?

(a) Prototyping

(b) Testing

(c) Empathizing ✓

(d) Defining

Correct answer: Empathizing

Multiple Choice Questions

8. What is the primary purpose of creating a business plan?

(a) To attract investors

(b) To outline strategies for growth 

(c) To study market trends

(d) To implement digital marketing strategies

Correct answer: To outline strategies for growth

Multiple Choice Questions

9. What does market analysis involve?

(a) Calculating revenue forecasts

(b) Researching competitors and understanding customer needs 

(c) Developing pricing strategies

(d) Implementing digital marketing campaigns

Correct answer: Researching competitors and understanding customer needs

Multiple Choice Questions

10. Which Sustainable Development Goal (SDG) focuses on affordable and clean energy?

(a) SDG5

(b) SDG7 

(c) SDG12

(d) SDG17

Correct answer: SDG7

Short Questions

1. What is the meaning of the word entrepreneur and its origin?

An entrepreneur is a person who starts and runs a new business or organization, taking risks and innovating in order to create value. The word comes from the French 'entreprendre', meaning 'to undertake': historically it described a person who undertakes a venture, accepting its risks in the hope of its rewards. Entrepreneurship is therefore the process of designing, launching and running such a venture, often initially a small business offering a product, process or service for sale or hire.

Short Questions

2. How do digital technologies help entrepreneurs reach a global audience?

Digital technologies remove the geographic limits of traditional business: a mobile app or website can be reached by customers anywhere in the world at any time, without opening shops in each country. Social media and online marketing let a small business advertise globally at low cost; e-commerce platforms handle orders, payments and delivery across borders; and digital tools provide translation, analytics and customer support that make serving distant customers practical. A student selling handmade crafts from one city can thus reach buyers on other continents, which a door-to-door or newspaper-advertising business never could.

Short Questions

3. What is an e-commerce platform, and why is it important for businesses?

An e-commerce platform is an online system (such as Shopify or Daraz) that lets a business display products, accept orders and receive payments over the Internet, without customers visiting a physical shop. It is important because it widens the market to anyone with Internet access, operates around the clock,

reduces the cost of a physical storefront, and provides built-in tools for inventory, payments, marketing and customer records. For small and new businesses it is often the difference between a local stall and a national or global shop.

Short Questions

4. Why is it important to identify market needs when generating a business idea?

Identifying market needs is important because a business succeeds only if it solves a real problem or satisfies a real want: an idea nobody needs will not sell, however clever it is. Researching needs — through surveys, market research and observation — reveals what customers lack, what competitors miss, and what they are willing to pay for, so the business idea can be shaped to a genuine gap in the market. It also reduces risk: the entrepreneur invests time and money in something evidence suggests people want, rather than guessing.

Short Questions

5. What is the main purpose of using design thinking in creative problem-solving?

The main purpose of design thinking is to solve problems from the user's point of view, ensuring the solution genuinely fits the people who will use it. Its five steps — empathizing (understanding users' real needs), defining (stating the problem clearly), ideating (generating many possible solutions), prototyping (building a simple version) and testing (trying it with users and improving it) — keep the process human-centred and iterative: instead of assuming what users want, the designer learns it, tries a solution, and refines it until it truly works. This reduces the risk of building an impressive product nobody needs.

Short Questions

6. Explain the importance of market analysis in a business plan. Give an example relevant to a local business in Pakistan.

Market analysis is important because it proves, with evidence, that a business has customers and a viable position: it researches competitors and understands customer needs, so the plan's sales forecasts, pricing and marketing rest on facts rather than hope. Without it, a business may enter a saturated market, misprice its product, or target customers who do not exist. Example: a student planning a small stationery shop near a school in Lahore would analyze how many students attend nearby schools, what existing shops charge for notebooks and pens, which items sell out fastest, and what students complain about (for example, high prices or poor quality). That analysis would guide the shop's location, prices, stock and opening offers — and would convince a lender or investor that the plan is realistic.

Short Questions

7. What are revenue models, and why are they essential components of a business plan?

A revenue model is the plan for how a business will generate income: what it charges, for what, and how often. Examples are direct sales (selling a product once), subscriptions (a monthly fee for continued service), advertising (free service funded by ads), and commission (a percentage of each transaction on a platform). Revenue models are essential in a business plan because they show how the business will sustain itself and become profitable: investors and lenders judge a plan largely by whether its revenue model is realistic, and the entrepreneur uses it to set prices, forecast income and decide when the business breaks even. A business with no clear revenue model has no credible path to survival.

Short Questions

8. How can businesses contribute to environmental sustainability according to the Sustainable Development Goals?

Businesses contribute to environmental sustainability by aligning their operations with the Sustainable Development Goals (SDGs), especially SDG 7 (affordable and clean energy), SDG 12 (responsible consumption and production) and SDG 13 (climate action). Practically, this means using clean energy (solar power for a shop or factory), reducing waste and packaging, recycling materials, using resources efficiently, and designing products that last and can be repaired. A business that does this cuts costs (less energy and waste), meets growing customer demand for responsible products, complies with environmental rules, and helps the country meet its SDG commitments — showing that profit and environmental responsibility can reinforce each other.

Long Questions

1. Explain why entrepreneurship is important for economic growth and innovation. Provide examples from the text to support your answer.

Entrepreneurship is important for economic growth because new businesses create jobs, increase competition and bring new products and services to market. When entrepreneurs start ventures — from a small online store to a technology company — they employ people, pay taxes, and force existing firms to improve, which raises overall productivity. It drives innovation because entrepreneurs spot unmet needs and take risks on new ideas that established firms avoid: digital entrepreneurs use mobile apps, e-commerce platforms and online marketing to reach global audiences, creating entirely new markets. In Pakistan, small digital businesses let students, artisans and farmers sell beyond their locality, spreading income more widely. Without entrepreneurship, economies stagnate: no new firms means no new jobs, no new ideas and no

competitive pressure to improve.

Long Questions

2. Explain how market research tools, online marketing tools, and e-commerce platforms can work together to help a business succeed in the digital world.

These three tool groups form a complete digital business loop. Market research tools (SurveyMonkey for surveys, and market research generally) discover what customers need and what competitors offer, so the business knows what to sell and at what price. Online marketing tools (such as Yoast SEO for search-engine optimization, Hootsuite for managing social media, and SEMrush for analysing search and competitors) make the business visible: SEO brings the shop to the top of search results, social media builds an audience, and analytics show which campaigns work. E-commerce platforms (such as Shopify) then convert that attention into sales by displaying products, processing orders and payments, and managing stock. Working together, research tells the business what to offer, marketing brings customers to it, and the platform turns visits into revenue — and the sales data feeds back into new research, improving the cycle.

Long Questions

3. Explain the design thinking process and how it can be applied to create a new product or service. Provide an example of how design thinking could be used to help farmers in rural Pakistan.

Design thinking is a five-step, human-centred process. (1) Empathizing: understand the users' real lives and problems through observation and interviews. (2) Defining: state the core problem clearly in one sentence. (3) Ideating: generate many possible solutions without judging them yet. (4) Prototyping: build a simple, cheap version of the best idea. (5) Testing: try it with real users, learn what fails, and improve it, repeating the loop. Example —

helping farmers in rural Pakistan: empathizing reveals that farmers lose crops because they lack timely weather and market-price information and often cannot read complex advisories. Defining gives the problem: 'Farmers need simple, timely crop advice in their own language.' Ideating produces options: an SMS advisory service, a voice-message helpline, a village noticeboard app. Prototyping builds a simple voice-message service in Urdu/Sindhi with daily weather and prices for one village. Testing with those farmers shows whether messages arrive, are understood and change decisions — and the service is refined (for example, adding pest alerts) before wider rollout. Design thinking thus ensures the product fits real users instead of assumptions.

Long Questions

4. Imagine you want to start a small online bookstore catering to students in your area. Develop a comprehensive business plan for your bookstore, including: (a) A market analysis focusing on student reading habits and competitors. (b) A revenue model outlining how you will generate income. (c) Digital marketing strategies you will use to promote your bookstore online. (Printed item 5, 'Ensure your answers are detailed and include practical examples relevant to starting your bookstore in Pakistan', is a direction for this question, not a separate question, and is followed here.)

(a) Market analysis: survey students in nearby schools and colleges about what they read (course books, exam guides, novels), how often they buy, and what they currently pay; study competitors — local bookshops (limited stock, fixed hours) and large online sellers (delivery charges, slow delivery to smaller towns) — and find the gap: affordable course books and popular novels delivered quickly and locally, with cash-on-delivery for students without bank cards. (b) Revenue model: primary income from direct sales with a margin on each book; additional income from a small delivery fee, a term-time subscription (monthly

book bundle at a discount), and commission on stationery sold alongside books; costs are stock, packaging, delivery and platform fees, with break-even when monthly sales cover rent-free, home-based overheads plus stock costs. (c) Digital marketing: create the store on an e-commerce platform (such as Shopify) optimized with Yoast SEO so searches for local course books find it; run social media pages showing new arrivals, student reviews and discount offers; use WhatsApp groups for schools and colleges; offer a first-order discount and a referral reward; and collect emails for exam-season promotions. Practical Pakistan-specific points: cash on delivery, Urdu/English listings, exam-season timing (admissions and board exams), and partnerships with school societies for promotion.

Long Questions

6. Choose a local business in Pakistan and discuss how it aligns with one or more Sustainable Development Goals (SDGs). Include its impact on social, environmental, and economic sustainability.

Example: a small solar-panel installation business in a Pakistani town. It aligns with SDG 7 (affordable and clean energy) by giving homes and shops cheaper, cleaner electricity than diesel generators or unreliable grid supply, and with SDG 8 (decent work and economic growth) by creating skilled local jobs for technicians and sales staff, and SDG 13 (climate action) by cutting carbon emissions. Social impact: families gain reliable light for children's study and clinics keep vaccines refrigerated, improving health and education. Environmental impact: each installation replaces fossil-fuel generation, reduces air pollution and waste from generator fuel, and the business can recycle old panels and batteries responsibly. Economic impact: customers save on electricity and fuel costs, local technicians earn steady incomes, and money stays in the local economy rather than going to imported fuel. The business shows how one

small venture can advance social, environmental and economic sustainability together.